

Creator Module — Product Requirements Document (PRD)

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1. Executive Summary

1.1 Purpose

The **Creator / Influencer Module** introduces a new user role (**Creator**) on the AOIN platform. Creators produce short-form video content (reels) for **merchant products**, upload them on AOIN, and earn **commission** on sales attributed to their reels. Merchants **hire** creators via **campaigns/deals**; AOIN tracks **attribution**, charges a **platform fee**, and manages **payouts**.

1.2 Goals

Stakeholder	Goal
Merchant	Outsource product marketing to creators; approve reels; pay commission on attributed sales.
Creator	Maintain portfolio; accept deals; upload campaign-linked reels; earn commission.
AOIN	Track reel→sale attribution; collect 5% platform fee on attributed sales; manage creator payouts.

1.3 Scope (V1)

- New role: `creator`
 - Creator onboarding (min 5 categories, availability, optional portfolio)
 - Campaign/Deal lifecycle (merchant creates → creator accepts → active)
 - Reel upload **linked to campaign** (creator path only for campaign reels)
 - **Last-touch attribution** (reel→product click within X days → purchase)
 - Settlement ledger (eligible earnings, AOIN fee, creator commission)
 - Merchant approval for reels before they go live
-

2. Current State Analysis

This section describes what exists in the Ecommerce_Backend codebase today, without the Creator module.

2.1 Roles & Authentication

Location: `auth/models/models.py`

- **User roles (enum `UserRole`):**
 - `USER` — Customer / end buyer
 - `MERCHANT` — Seller; has `MerchantProfile`
 - `ADMIN` — Platform admin
 - `SUPER_ADMIN` — Super admin
- **No `CREATOR` role exists.** Creators are not represented in the system.
- **User model:** `id`, `email`, `password_hash`, `first_name`, `last_name`, `phone`, `role`, `is_active`, `is_email_verified`, `is_phone_verified`, `auth_provider`, etc.
- **MerchantProfile:** One-to-one with `User` when `role == MERCHANT`. Holds business details, verification status, subscription, bank/ShipRocket fields. **No creator-specific profile exists.**

2.2 Reels (Current Flow)

Location: `models/reel.py`, `controllers/reels_controller.py`, `routes/reels_routes.py`

Current reel model (relevant fields):

Field	Type	Notes
<code>reel_id</code>	PK	
<code>merchant_id</code>	FK → <code>merchant_profiles</code>	Required. Only merchants can own reels.
<code>product_id</code>	FK → <code>products</code> , nullable	AOIN product or NULL for “external” reels.
<code>product_url</code> , <code>product_name</code> , <code>category_id</code> ,	Various	For external reels when <code>product_id</code> is NULL.

<code>category_name</code> , <code>platform</code>		
<code>video_url</code> , <code>thumbnail_url</code> , <code>description</code> , stats, etc.	—	Standard reel content.
<code>approval_status</code>	String	Default <code>'approved'</code> ; reels are effectively live without merchant approval flow.
<code>approved_by</code>	FK → users	Optional.

Current reel upload flow:

1. **Who can upload:** Only users with a **MerchantProfile** (i.e. merchants). Controllers explicitly reject non-merchants with: *"Only merchants can upload reels"*.
2. **Product link:** Either **AOIN product** (`product_id` + validation that product belongs to merchant and is approved/stocked) or **external** (`product_url`, `product_name`, etc.).
3. **No campaign:** There is no `campaign_id`, deal, or creator. Reels are **merchant-owned** and tied only to merchant + product (or external link).
4. **Visibility:** Reels are shown in feed; visibility logic uses `get_visible_reels()` (product stock, approval, etc.). No concept of "pending merchant approval" for a creator-submitted reel.

Conclusion: Today, reels are a **merchant-only** feature. There is no creator path, no campaign, and no attribution.

2.3 Products & Categories

Location: `models/product.py`, `models/category.py`

- **Product:** `product_id`, `merchant_id`, `category_id`, `brand_id`, name, pricing, approval, stock (via `ProductStock`), etc.
- **Category:** `category_id`, `parent_id`, `name`, `slug`, etc. Used for product taxonomy and filtering.
- **No creator–category association:** Categories are not linked to creators (e.g. "creator's chosen categories for reels").

2.4 Orders & Settlement

Location: `models/order.py`, `models/shipment.py`, `models/enums.py`

- **Order:** `order_id`, `user_id`, `order_status`, amounts, payment fields, addresses. Status enum includes: `PENDING_PAYMENT`, `PROCESSING`, `SHIPPED`, `DELIVERED`, `CANCELLED_*`, `REFUNDED`, `RETURN_*`, etc.
- **OrderItem:** `order_item_id`, `order_id`, `product_id`, `merchant_id`, quantity, prices, `item_status`, discount fields. **No attribution field** (no `campaign_id`, `reel_id`, or `creator_id`).
- **Shipment:** Tied to order and merchant; has `actual_delivery_date`, `shipment_status`.
- **Eligibility:** Delivery and return/refund flows exist in enums and status history, but there is **no settlement/ledger or creator payout** logic.

Conclusion: Orders and order items are not linked to reels or campaigns. No attribution, no commission calculation, no creator earnings.

2.5 Visit / Tracking

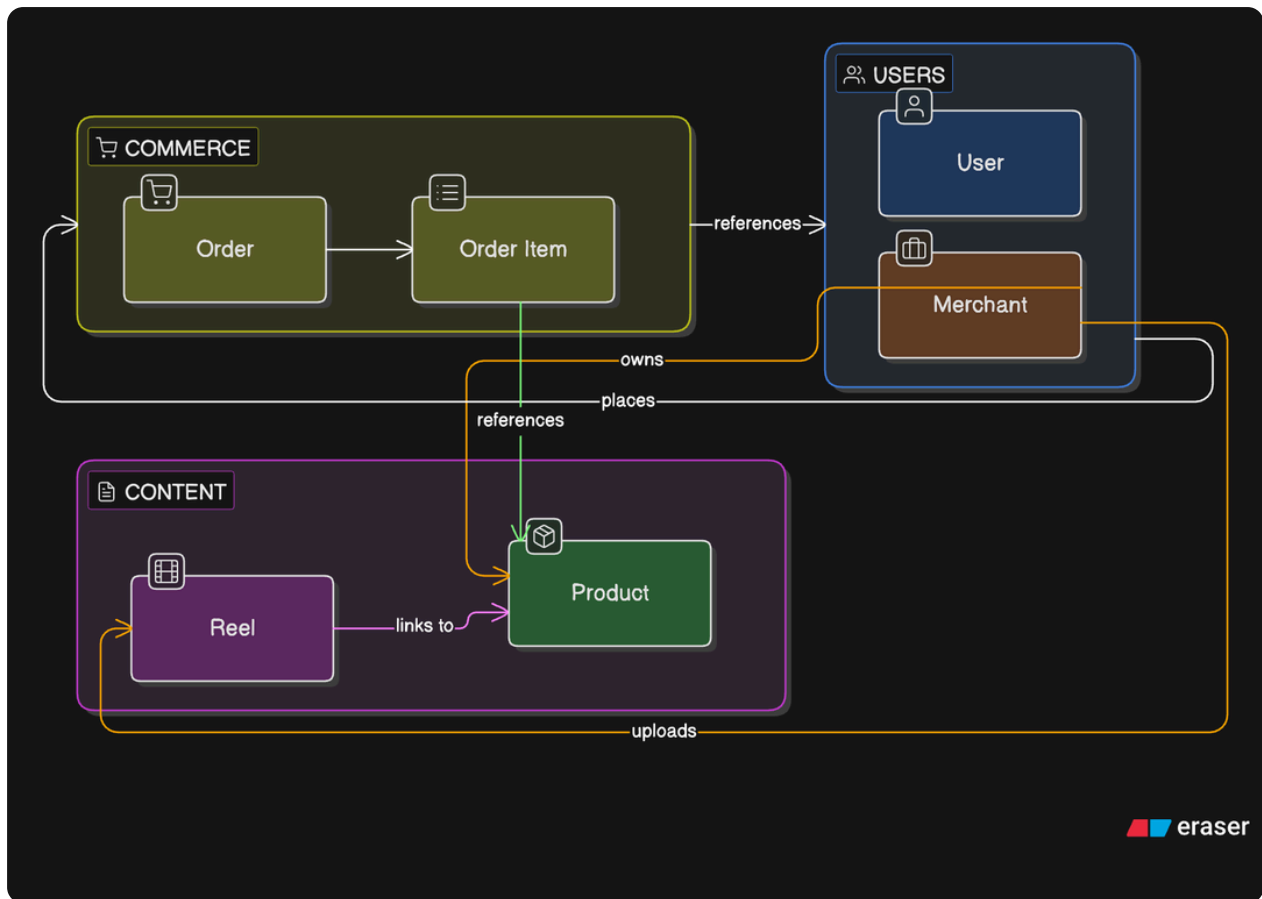
Location: `models/visit_tracking.py`

- **VisitTracking:** Session, IP, landing page, referrer, device, `user_id` (optional). Used for generic visit/session analytics and conversion (e.g. signup). **No reel view or “reel→product click” event.** No attribution pipeline.

2.6 Current High-Level Flow (As-Is)

🔴 Mermaid Diagram

```
flowchart LR
    subgraph Users
        U[User/Customer]
        M[Merchant]
    end
    subgraph Content
        P[Product]
        R[Reel]
    end
    subgraph Commerce
        O[Order]
        OI[Order Item]
    end
    M -->|owns| P
    M -->|uploads| R
    R -->|links to| P
    U -->|places| O
    O --> OI
    OI -->|references| P
    OI -->|references| M
```



- **Reel path:** Merchant only → upload reel → link to own product (or external). No creator, no campaign.
- **Purchase path:** User browses (including reels), places order; order items reference product and merchant only. No attribution to any reel or creator.

3. Target State & Requirements

3.1 Roles (Target)

Role	Description
User/Customer	Views reels, opens products, purchases.
Merchant	Lists products; hires creators (campaigns); approves reels; pays commission on attributed sales.
Creator/Influencer	NEW. Maintains portfolio & categories; accepts deals; uploads campaign-linked reels only.
Admin / Super Admin	Moderation, disputes, overrides (optional in V1).

3.2 Creator Lifecycle (Target)

- **Signup / Login:** Same auth as today; new role value `creator`. No MerchantProfile; creator has a **CreatorProfile** (or equivalent).

- **Onboarding (mandatory):**
 - Select **at least 5 categories** in which they create reels.
 - Optional: language, portfolio links, sample reels.
 - **Availability:** Available / Busy.
- **Dashboard (V1):** Home (active deals, pending approvals, earnings, deadlines), My Categories, Portfolio, Deals/Campaigns (offers, active, completed), **Upload Reel** (only for assigned active campaigns), Earnings & Payouts, Settings.

3.3 Merchant Lifecycle (Target)

- **Creator Marketing section:** Find creators (filter by category, etc.), create campaign (offer), manage campaigns (approve reels, see attributed sales), payout/commission summary per creator.
- **Hire flow:** Select product → system suggests creators by **category match** + portfolio + availability → merchant selects creator and creates **Deal/Campaign** (product_id, creator_id, commission %, optional cap, campaign window, deliverable e.g. "1 reel") → creator accepts → campaign becomes **Active**.

3.4 Campaign / Deal Lifecycle (Target)

- **Statuses (V1):** Draft → Sent → Accepted → Active → Submitted → Approved → Live → Completed.
- **Side states:** Rejected (creator rejects), Cancelled/Expired (merchant/admin or timeout).
- **Identification:** Internal `campaign_id`; short human-friendly `campaign_code` for display and optional lookup.

3.5 Reel Upload (Creator Path, Target)

- Creator can upload reels **only under assigned active campaigns**.
- Upload UI: **Select Campaign** (dropdown of active campaigns for that creator), upload video, description, submit.
- **Merchant approval:** Reel does not go to public feed until merchant approves (wrong product, quality, compliance).
- Backend: Reel record stores **campaign_id**; campaign already has product_id, merchant_id, creator_id. Chain: **Reel → Campaign → Product → Merchant + Creator**.

3.6 Attribution (Target)

- **Rule (V1): Last-touch attribution.** User sees reel → clicks to product → purchases. If purchase occurs **within X days** (e.g. 7) of the **last** reel→product click, that sale is attributed to the reel's campaign (and thus creator).
- **Implementation:** Store **reel→product click events** (user, reel_id, product_id, campaign_id, timestamp). On order placement (or on delivery/eligibility), resolve last touch within window and set **attributed_campaign_id** (and hence creator) per order item or order.

3.7 Commission & Payout (Target)

- **Deal terms (V1):** (1) **Percent + cap** (e.g. 20% up to 200 qty), or (2) **Percent unlimited** (e.g. 10% on all attributed sales).
 - **AOIN fee:** 5% on attributed sales (campaign-attributed only).
 - **Commission base:** Net = item price – item discount – refunds; exclude shipping/tax.
 - **Eligibility:** Order delivered + return/refund window passed (e.g. 7/14 days) → sale eligible for settlement.
 - **Ledger:** Creator earning, AOIN fee, merchant net (informational). Payout: weekly/monthly or threshold-based. Refund/return → reversal/adjustment of creator payout and AOIN fee.
-

4. Gap Analysis

Area	Current State	Target State	Gap
Roles	USER, MERCHANT, ADMIN, SUPER_ADMIN	+ CREATOR	Add <code>UserRole.CREATOR</code> and creator-specific profile/onboarding.
Reels	Merchant-only upload; no campaign	Creator upload for assigned campaigns; reel has <code>campaign_id</code>	New creator upload path; add <code>campaign_id</code> to reels; merchant approval workflow for creator reels.
Campaign/Deal	Does not exist	Full lifecycle (Draft→... →Completed); terms, code	New Campaign/Deal model and APIs (create, list, accept, reject, status).
Creator profile	Does not exist	Categories (≥5), portfolio, availability	New CreatorProfile (or equivalent) + onboarding APIs.
Attribution	None	Last-touch; reel→product click events	New attribution event model + click logging API + attribution resolution on order/delivery.
Order/OrderItem	No link to reel/campaign	Attributed sales per campaign/creator	Add attribution fields (e.g. <code>attributed_campaign_id</code>) and settlement logic.
Settlement/Payout	None	Ledger, eligibility, payouts, reversals	New ledger/earnings model + eligibility rules + payout flow (V1 can be manual or batch).

5. Business Logic Flows & Diagrams

5.1 Role & Actor Overview

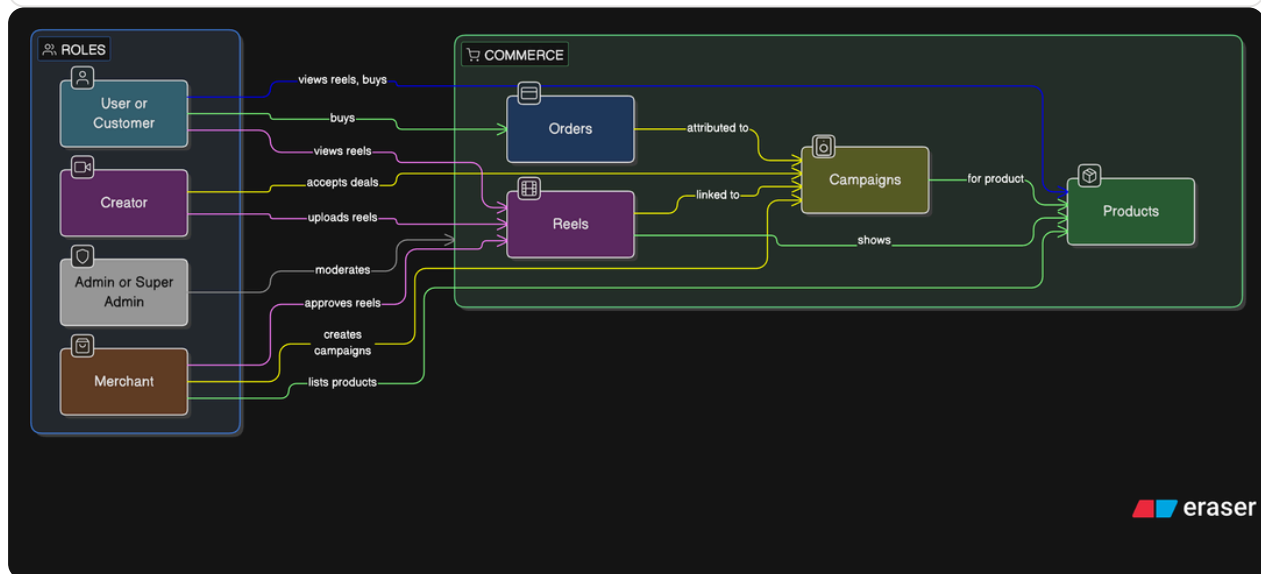
Mermaid Diagram

```
graph TD
    subgraph Roles
        U[User/Customer]
        M[Merchant]
        C[Creator]
        A[Admin/Super Admin]
    end

    subgraph Commerce
        Prod[Products]
        Camp[Campaigns]
        Reel[Reels]
        Ord[Orders]
    end

    U -->|views reels, buys| Commerce
    M -->|lists products, creates campaigns, approves reels| Commerce
    C -->|accepts deals, uploads reels| Commerce
    A -->|moderates| Commerce

    Camp -->|for product| Prod
    Reel -->|linked to| Camp
    Reel -->|shows| Prod
    Ord -->|attributed to| Camp
```



5.2 Campaign Lifecycle (State Machine)

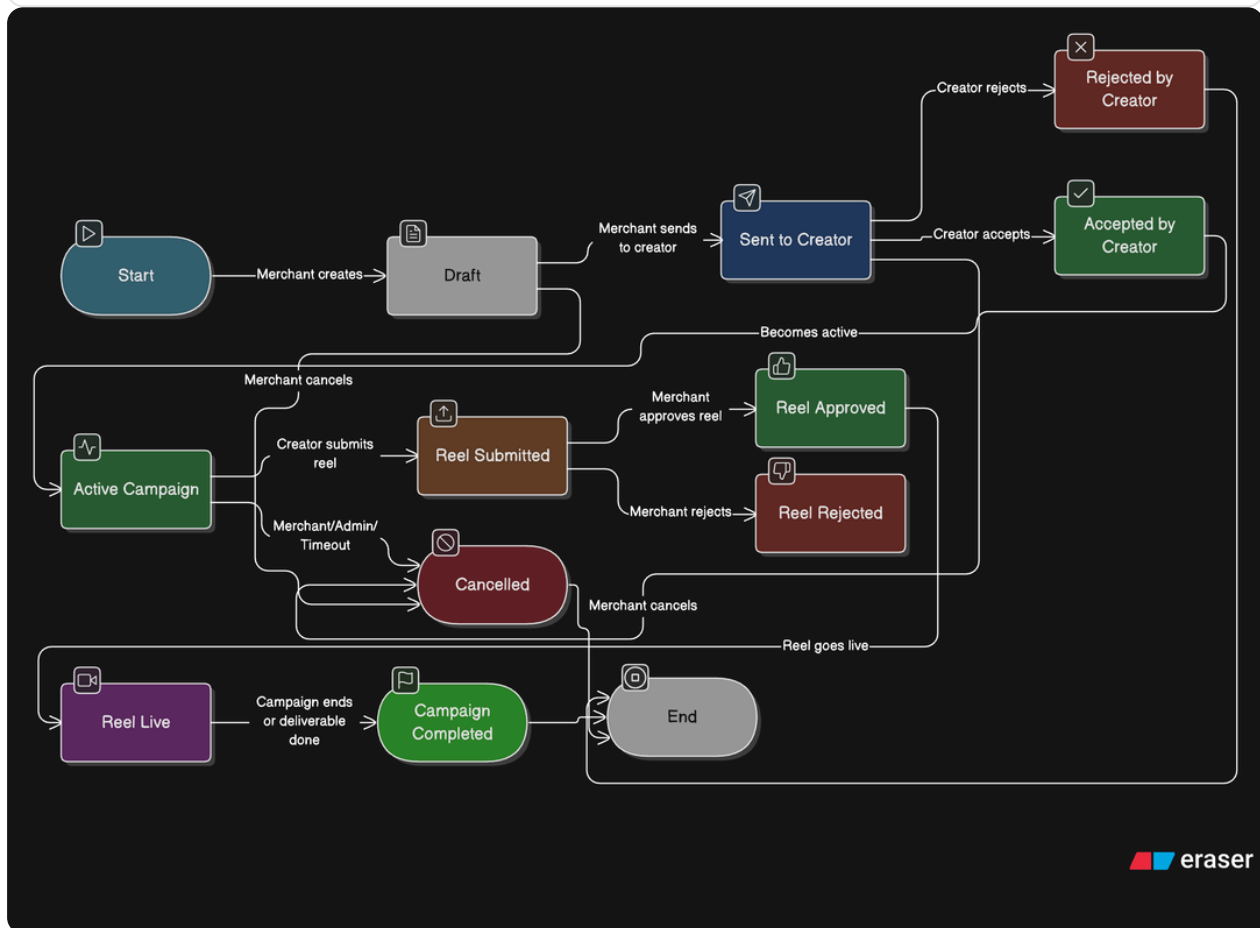
Mermaid Diagram

stateDiagram-v2

```

[*] --> Draft: Merchant creates
Draft --> Sent: Merchant sends to creator
Sent --> Accepted: Creator accepts
Sent --> Rejected: Creator rejects
Accepted --> Active: Becomes active
Active --> Submitted: Creator submits reel
Submitted --> Approved: Merchant approves reel
Submitted --> Rejected: Merchant rejects
Approved --> Live: Reel goes live
Live --> Completed: Campaign ends / deliverable done
Draft --> Cancelled: Merchant cancels
Sent --> Cancelled: Merchant cancels
Active --> Cancelled: Merchant/Admin/Timeout
Rejected --> [*]
Cancelled --> [*]
Completed --> [*]

```

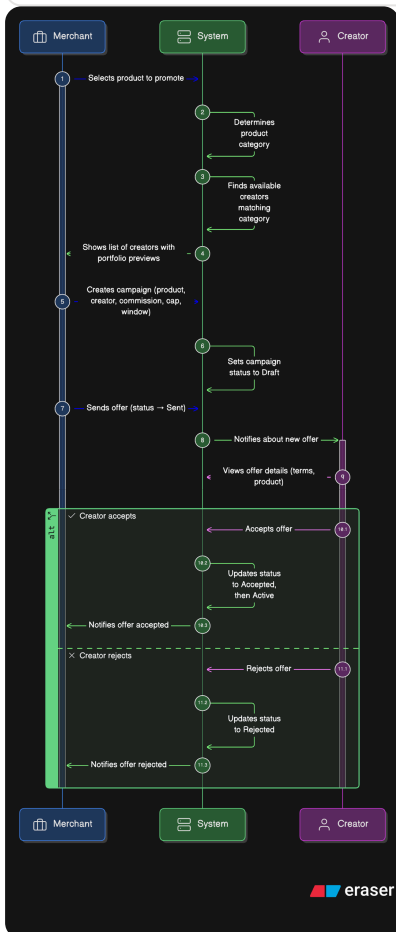


5.3 Merchant: Create Campaign & Creator Accepts

Mermaid Diagram

sequenceDiagram

```
participant Merchant
participant System
participant Creator
Merchant->>System: Select product (AOIN)
System->>System: Get product category
System->>System: Find creators: category match + availability
System->>Merchant: List creators (portfolio preview)
Merchant->>System: Create campaign (product_id, creator_id, commission %,
cap?, window)
System->>System: Campaign status = Draft
Merchant->>System: Send offer (status → Sent)
System->>Creator: Notify: New offer
Creator->>System: View offer (terms, product)
alt Creator accepts
Creator->>System: Accept
System->>System: Status → Accepted → Active
System->>Merchant: Notify: Accepted
else Creator rejects
Creator->>System: Reject
System->>System: Status = Rejected
System->>Merchant: Notify: Rejected
end
end
```



5.4 Creator: Upload Reel (Campaign-Linked)

Mermaid Diagram

sequenceDiagram

participant Creator

participant System

participant Merchant

Creator->>System: Get my active campaigns (dropdown)

System->>Creator: List campaigns (campaign_code, product, terms)

Creator->>System: Upload reel (campaign_id, video, description)

System->>System: Validate: campaign active, creator assigned

System->>System: Create reel (campaign_id, merchant_id from campaign, product_id from campaign)

System->>System: Reel approval_status = pending (creator reel)

System->>System: Campaign status → Submitted (if first reel) or keep Active

System->>Merchant: Notify: Reel submitted for approval

Merchant->>System: View pending reels

alt Approve

Merchant->>System: Approve reel

System->>System: approval_status = approved, reel goes live

System->>System: Campaign → Live (or stay Active for more reels)

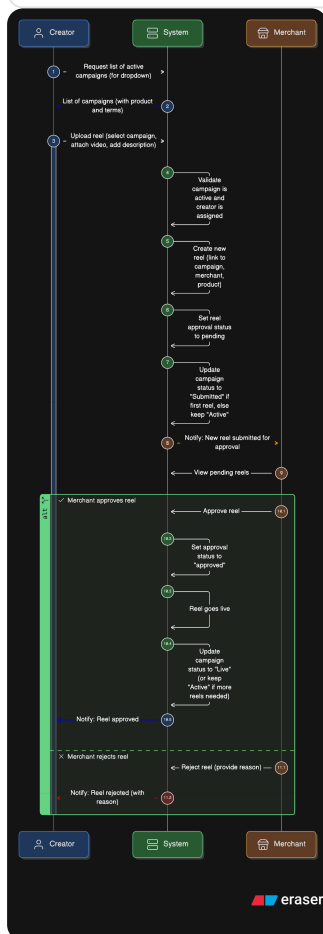
System->>Creator: Notify: Reel approved

else Reject

Merchant->>System: Reject (reason)

System->>Creator: Notify: Reel rejected

end

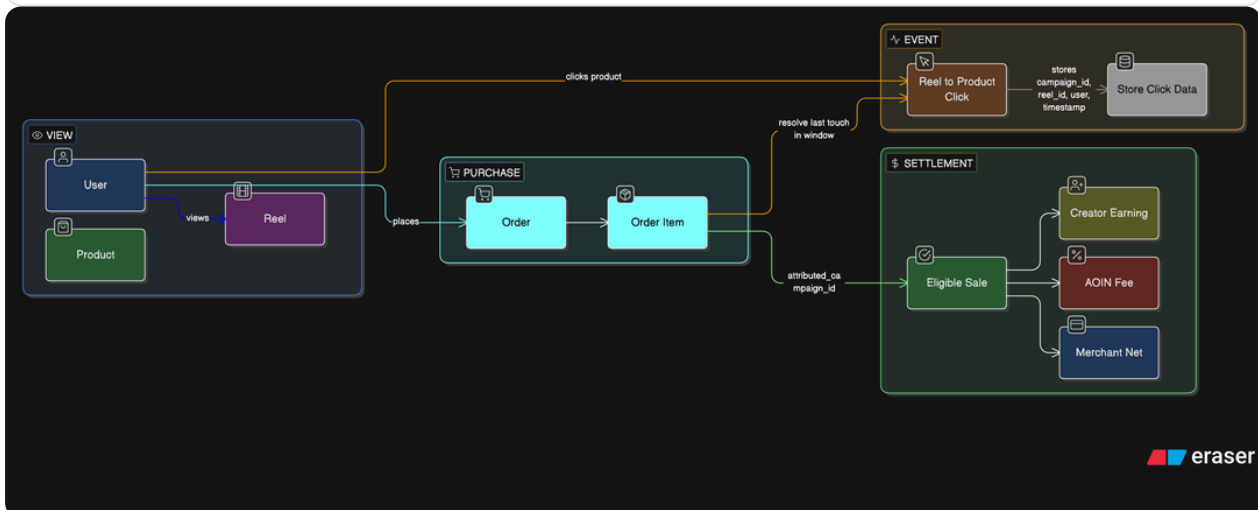


5.5 Attribution: Reel Click → Purchase → Settlement

Mermaid Diagram

```

flowchart LR
    subgraph View
        U[User]
        R[Reel]
        P[Product]
    end
    end
    subgraph Event
        E[Reel→Product Click]
    end
    end
    subgraph Purchase
        O[Order]
        OI[Order Item]
    end
    end
    subgraph Settlement
        EL[Eligible Sale]
        CR[Creator Earning]
        AF[AOIN Fee]
        MN[Merchant Net]
    end
    end
    U -->|views| R
    U -->|clicks product| E
    E -->|stores campaign_id, reel_id, user, ts| DB
    U -->|places| O
    O --> OI
    OI -->|resolve last touch in window| E
    OI -->|attributed_campaign_id| EL
    EL --> CR
    EL --> AF
    EL --> MN
  
```

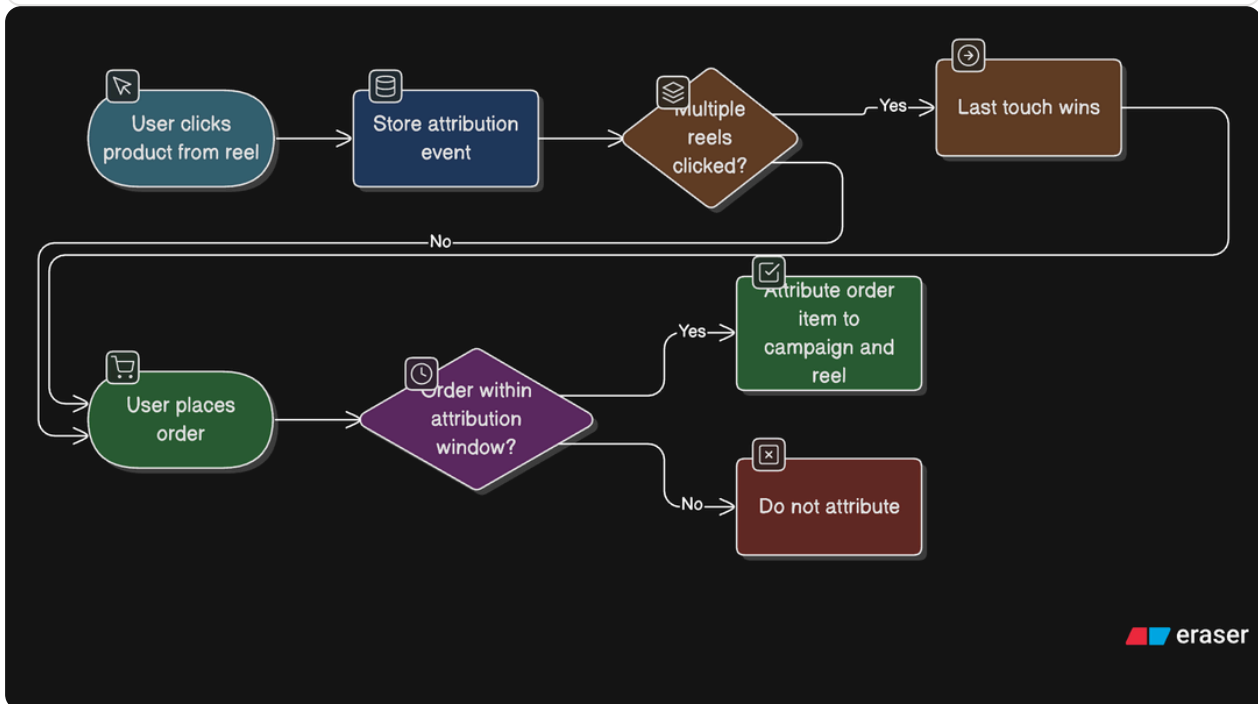


5.6 Last-Touch Attribution Rule (Conceptual)

Mermaid Diagram

flowchart TB

```
A[User clicks product from reel at T0] --> B[Store attribution event:
user_id, reel_id, campaign_id, product_id, T0]
B --> C[User places order at T1] --> D{T1 - T0 <= X days?}
D -->|Yes| E[Attribute order item to campaign/reel]
D -->|No| F[Do not attribute]
G[Multiple reels clicked?] --> H[Last touch wins]
H --> D
```



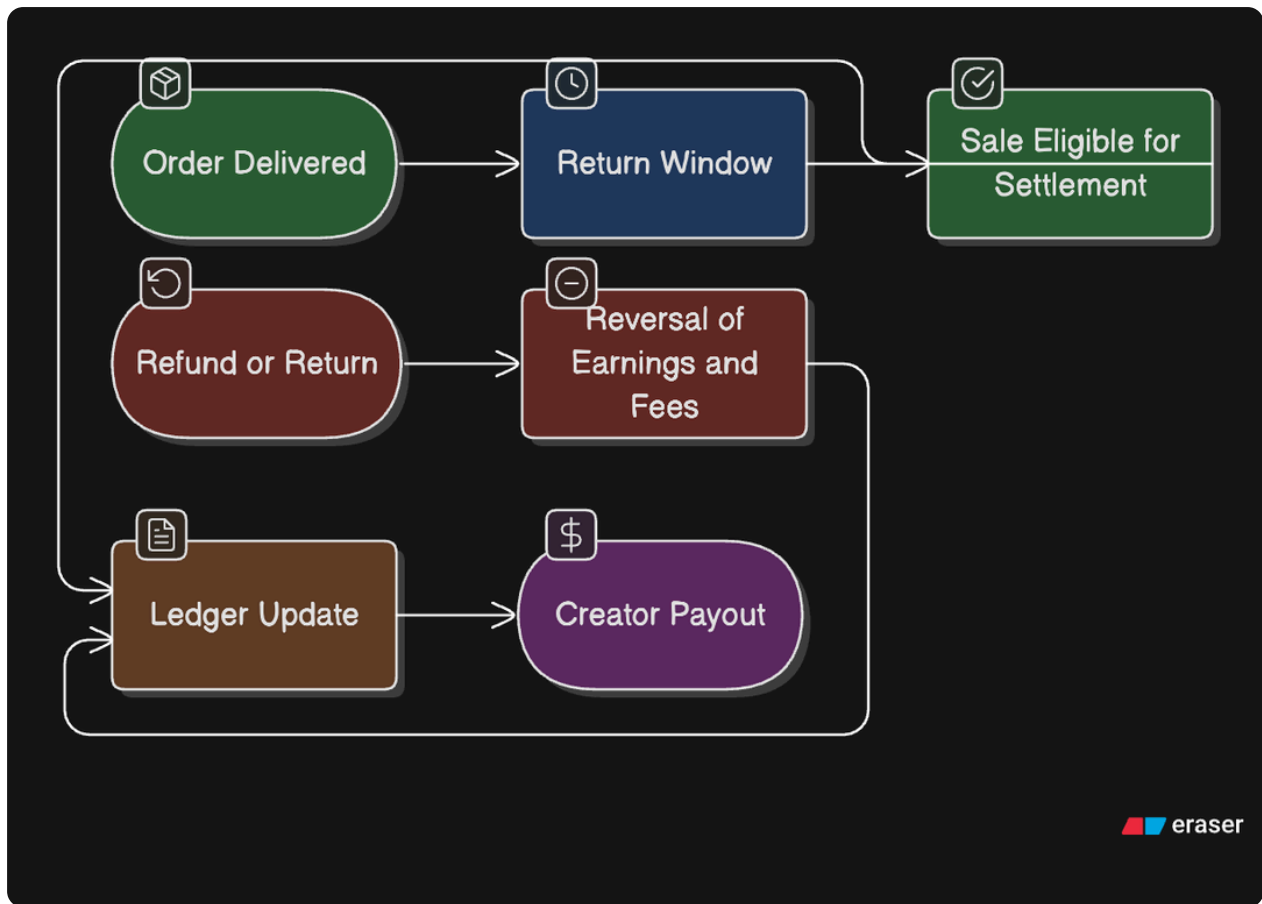
eraser

5.7 Settlement & Payout Flow (Conceptual)

Mermaid Diagram

flowchart TB

```
O[Order delivered] --> W[Return window e.g. 7/14 days]
W --> EL[Sale eligible for settlement]
EL --> L[Ledger: Creator earning, AOIN fee, Merchant net]
L --> P[Creator payout: weekly/monthly or threshold]
R[Refund/Return] --> RV[Reversal: creator + AOIN fee]
RV --> L
```



5.8 Current vs Target: Reel Ownership & Upload Paths

Current (reels are merchant-only):

Y Mermaid Diagram

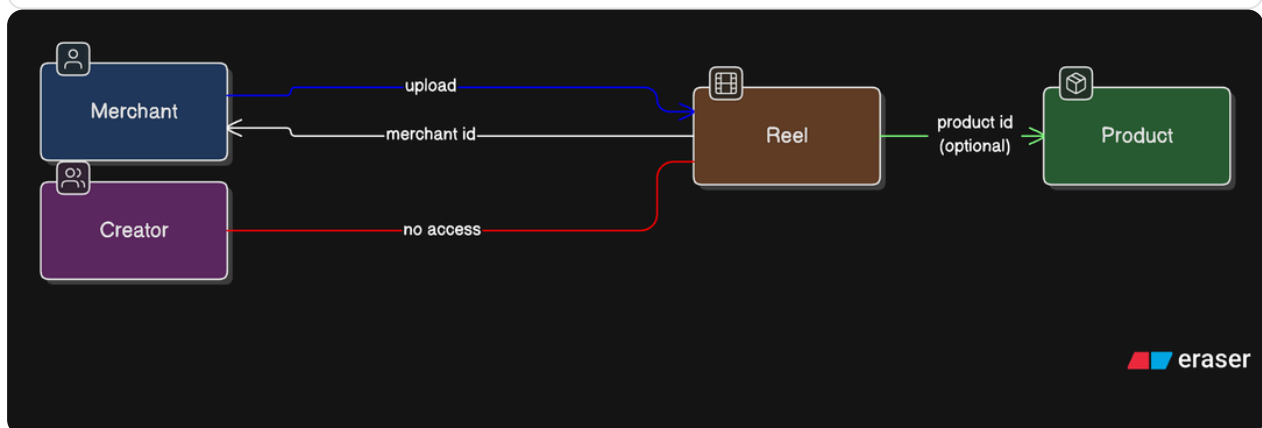
flowchart LR

M[Merchant] -->|upload| R[Reel]

R -->|merchant_id| M

R -->|product_id optional| P[Product]

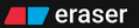
C[Creator] -. ->|no access| R



Target (two paths: merchant direct + creator via campaign):

flowchart LR

```
M[Merchant] -->|upload own reels| R1[Reel - merchant]
R1 -->|merchant_id| M
R1 -->|product_id| P[Product]
Camp[Campaign] -->|product_id, merchant_id| P
Camp -->|assigned to| C[Creator]
C -->|upload for campaign| R2[Reel - creator]
R2 -->|campaign_id| Camp
R2 -->|merchant_id from campaign| M
R2 -->|product_id from campaign| P
M -->|approve| R2
```



6. UI/UX Sections & Screen Maps

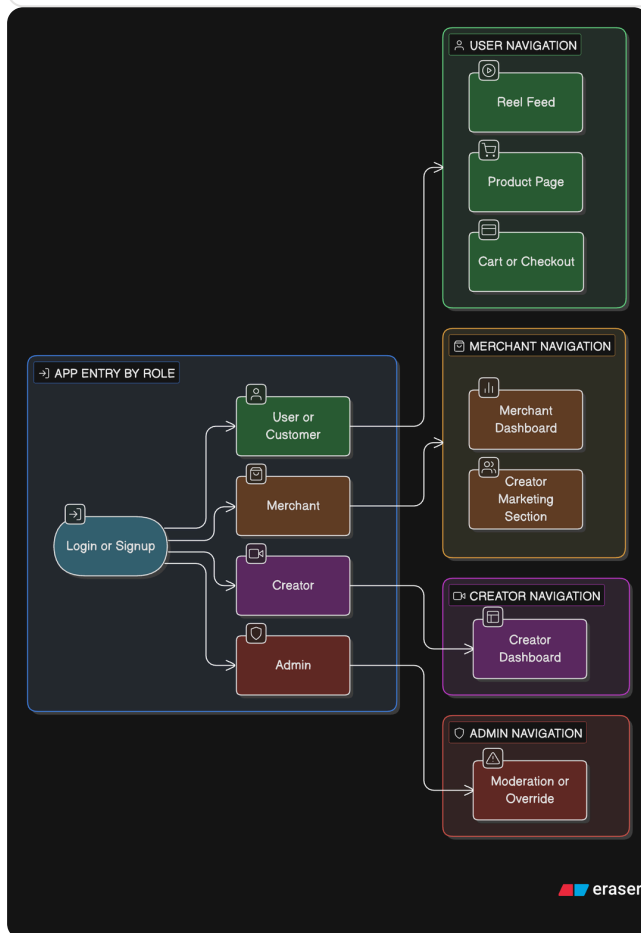
This section describes every possible UI section and screen for the Creator module, by role. Use these as a reference for wireframes, navigation, and UX flows.

6.1 Overall: Who Sees What (Role-Based Entry)

Mermaid Diagram

```
graph TD
    subgraph Entry ["App Entry by Role"]
        L[Login / Signup] --> U[User/Customer]
        L --> M[Merchant]
        L --> C[Creator]
        L --> A[Admin]
    end

    U --> U_Reel[Reel Feed]
    U --> U_Product[Product Page]
    U --> U_Cart[Cart / Checkout]
    M --> M_Dash[Merchant Dashboard]
    M --> M_Creator[Creator Marketing Section]
    C --> C_Dash[Creator Dashboard]
    A --> A_Mod[Moderation / Override]
```



6.2 Creator Dashboard — Section Hierarchy

🐙 Mermaid Diagram

flowchart TB

```
subgraph Creator_Dashboard["Creator Dashboard (post-login)"]
  Nav[Sidebar / Bottom Nav]
  Nav --> Home[Home / Overview]
  Nav --> Categories[My Categories]
  Nav --> Portfolio[Portfolio]
  Nav --> Deals[Deals / Campaigns]
  Nav --> Upload[Upload Reel]
  Nav --> Earnings[Earnings & Payouts]
  Nav --> Settings[Settings]
end

Home --> H1[Active deals count]
Home --> H2[Pending approvals count]
Home --> H3[Earnings summary]
Home --> H4[Upcoming deadlines]

Categories --> Cat1[Selected 5+ categories list]
Categories --> Cat2[Edit categories - min 5]

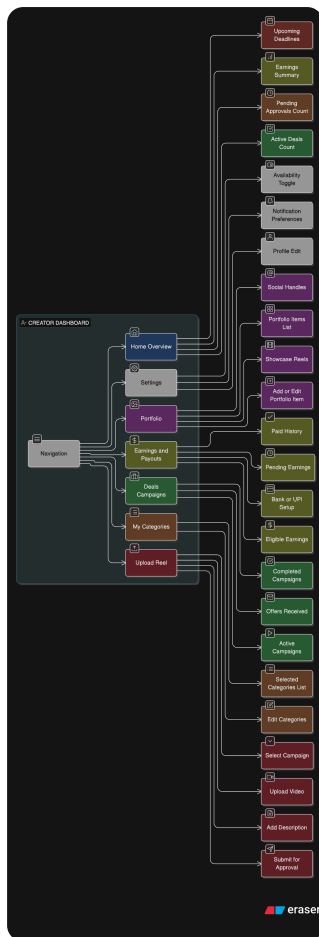
Portfolio --> P1[Portfolio items list]
Portfolio --> P2[Links / showcase reels]
Portfolio --> P3[Social handles]
Portfolio --> P4[Add / Edit portfolio item]

Deals --> D1[Offers received - Accept/Reject]
Deals --> D2[Active campaigns - Submit reel, deadline]
Deals --> D3[Completed campaigns - History]

Upload --> U1[Select Campaign dropdown]
Upload --> U2[Upload video]
Upload --> U3[Add description]
Upload --> U4[Submit for approval]

Earnings --> E1[Pending earnings]
Earnings --> E2[Eligible earnings]
Earnings --> E3[Paid history]
Earnings --> E4[Bank/UPI setup - when payout live]

Settings --> S1[Profile edit]
Settings --> S2[Availability toggle]
Settings --> S3[Notification preferences]
```



6.3 Creator: Screen Flow (User Journey)

Y Mermaid Diagram

flowchart LR

subgraph Onboarding

O1[Signup as Creator]

O2[Onboarding: 5+ categories]

O3[Availability / Portfolio optional]

O1 --> O2 --> O3

end

subgraph Daily

D1[Home - see offers & stats]

D2[Deals - accept/reject offer]

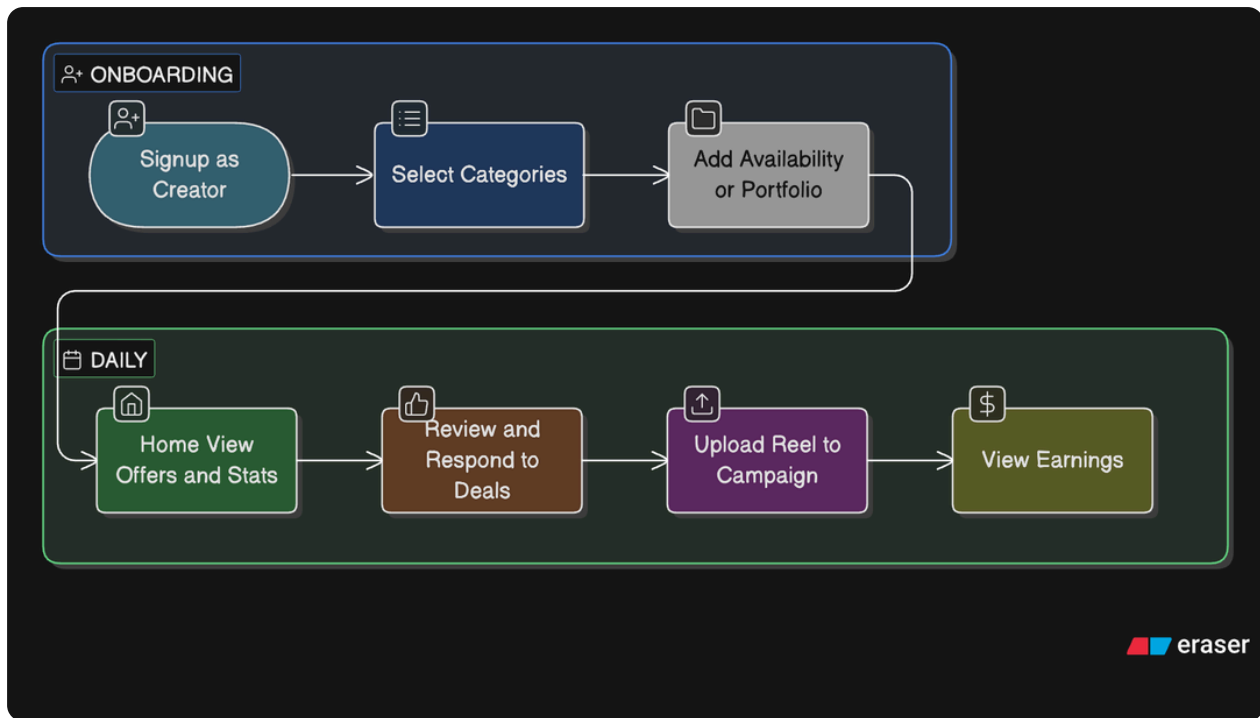
D3[Upload Reel - select campaign, upload]

D4[Earnings - view pending/paid]

D1 --> D2 --> D3 --> D4

end

Onboarding --> Daily



6.4 Merchant Dashboard — Creator Marketing Section

Y Mermaid Diagram

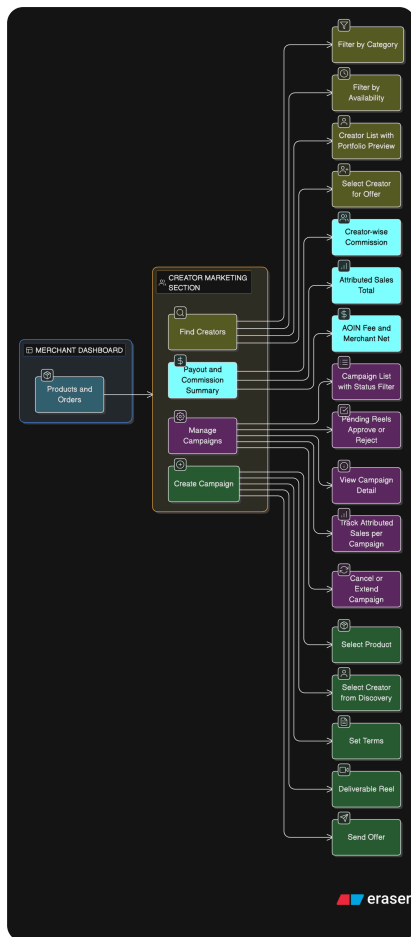
flowchart TB

```

subgraph Merchant_Dashboard["Merchant Dashboard"]
    Main[Existing: Products, Orders, etc.]
    CreatorSection[Creator Marketing / Creator Campaigns]
    Main --> CreatorSection
end

subgraph Creator_Marketing["Creator Marketing Section"]
    Find[Find Creators]
    Create[Create Campaign]
    Manage[Manage Campaigns]
    PayoutSummary[Payout / Commission Summary]
end

Find --> F1[Filter by category]
Find --> F2[Filter by availability]
Find --> F3[Creator list with portfolio preview]
Find --> F4[Select creator for offer]
Create --> C1[Select product - AOIN product]
Create --> C2[Select creator - from discovery]
Create --> C3[Set terms: commission %, cap, window]
Create --> C4[Deliverable: 1 reel - V1]
Create --> C5[Send offer]
Manage --> M1[Campaign list - status filter]
Manage --> M2[View campaign detail - terms, creator]
Manage --> M3[Pending reels - Approve / Reject]
Manage --> M4[Track attributed sales per campaign]
Manage --> M5[Cancel / Extend campaign]
PayoutSummary --> PS1[Creator-wise commission]
PayoutSummary --> PS2[Attributed sales total]
PayoutSummary --> PS3[AOIN fee / Merchant net]
  
```



6.5 Merchant: Creator Campaign User Journey

Mermaid Diagram

flowchart LR

subgraph Create_Flow

A1[Select product]

A2[Find creators - category match]

A3[Choose creator]

A4[Create campaign - terms]

A5[Send offer]

A1 --> A2 --> A3 --> A4 --> A5

end

subgraph Manage_Flow

B1[See accepted campaign]

B2[Reel submitted - notify]

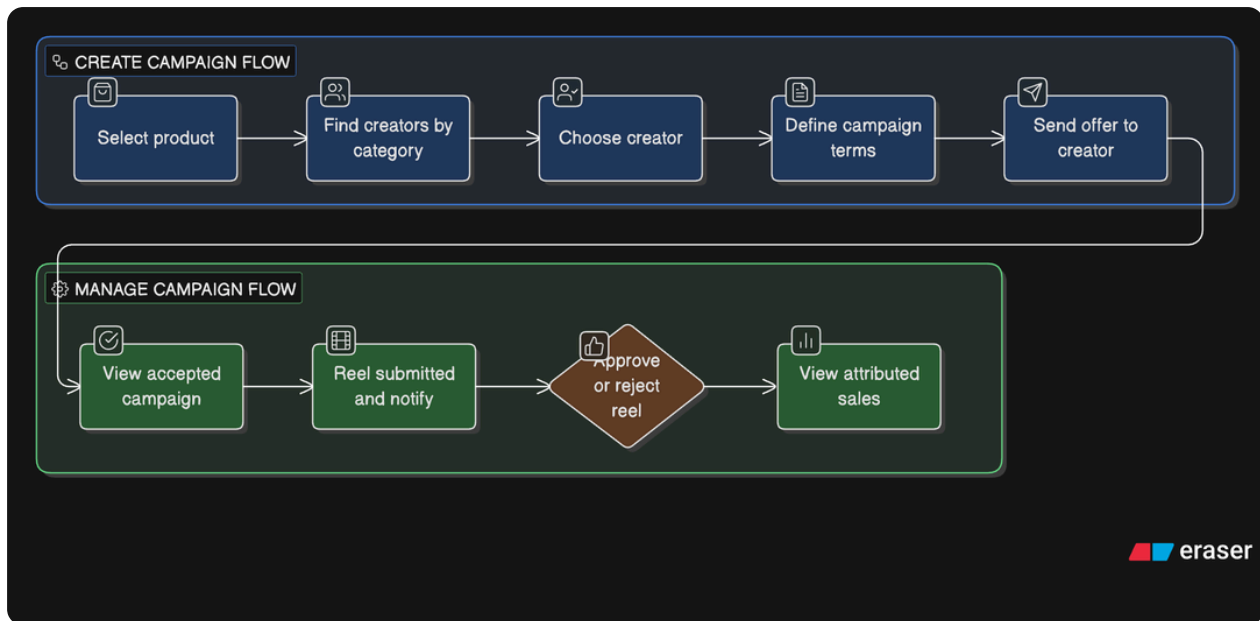
B3[Approve / Reject reel]

B4[View attributed sales]

B1 --> B2 --> B3 --> B4

end

Create_Flow --> B1



6.6 Customer / User: Reel & Product (Attribution UX)

Mermaid Diagram

flowchart TB

```

    subgraph Reel_Feed["Reel Feed (existing + creator reels)"]
        RF1[Vertical reel feed]
        RF2[Reel card: video, description]
        RF3[Product CTA - link to product]
        RF1 --> RF2 --> RF3
    end

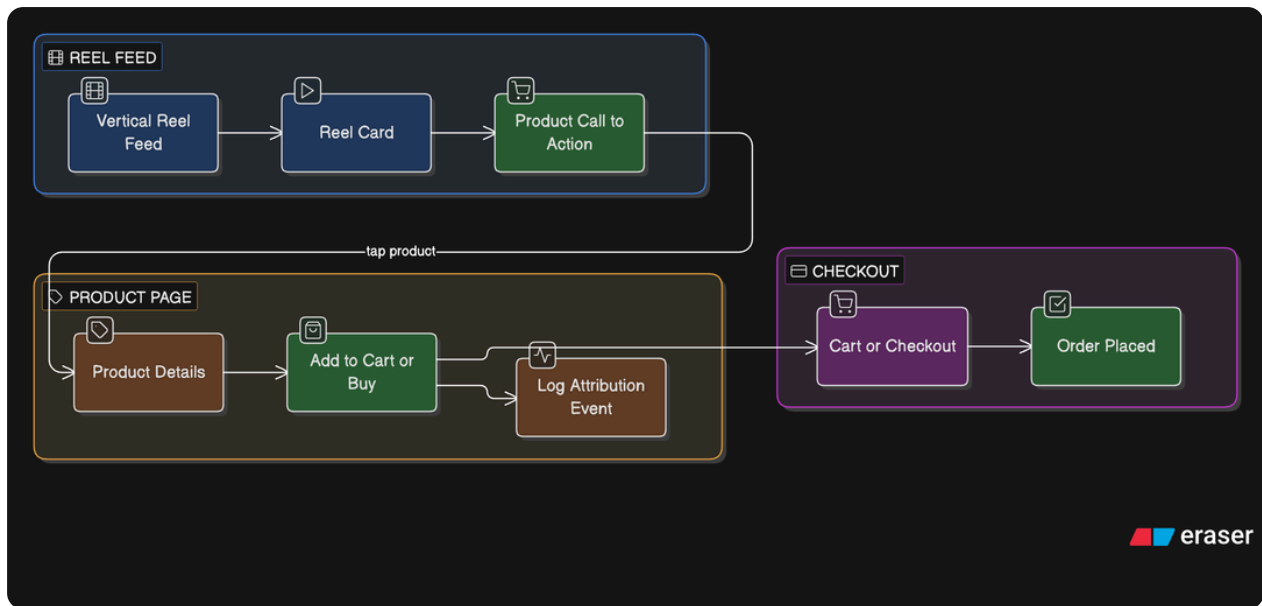
    end

    subgraph Product_Page["Product Page (attribution)"]
        PP1[Product details]
        PP2[Add to cart / Buy]
        PP3[Backend: log reel→product click - attribution event]
        PP1 --> PP2
        RF3 -->|tap product| PP1
        PP2 --> PP3
    end

    end

    subgraph Checkout
        CH1[Cart / Checkout]
        CH2[Order placed - last-touch attribution resolved]
        PP2 --> CH1 --> CH2
    end

    end
  
```



6.7 Creator Onboarding — Screen Breakdown

Y Mermaid Diagram

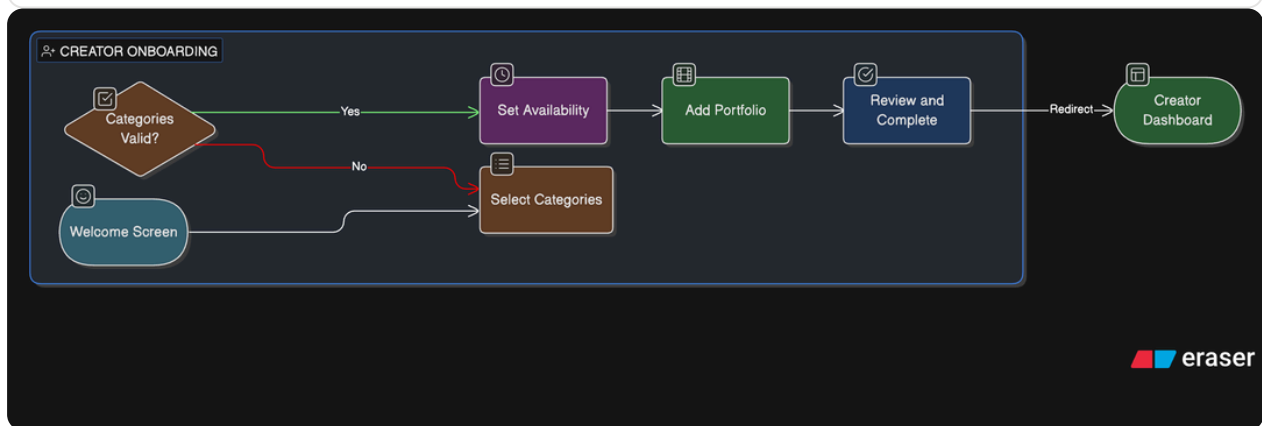
flowchart TB

```

subgraph Onboarding_Screens["Creator Onboarding Screens"]
    S1[Welcome - You are signing up as Creator]
    S2[Categories - Select min 5 categories]
    S3[Availability - Available / Busy toggle]
    S4[Portfolio - Optional: links, sample reels]
    S5[Review & Complete]
    S1 --> S2 --> S3 --> S4 --> S5
end

S2 --> |validation| S2
S5 --> |redirect| Dashboard[Creator Dashboard]

```



6.8 Campaign Lifecycle — UI States per Screen

Mermaid Diagram

stateDiagram-v2

direction LR

[*] --> Draft: Merchant: Create

Draft --> Sent: Merchant: Send

Sent --> OffersScreen: Creator: See offers

OffersScreen --> Accepted: Creator: Accept

OffersScreen --> Rejected: Creator: Reject

Accepted --> ActiveScreen: Creator: Active campaigns

ActiveScreen --> UploadScreen: Creator: Upload reel

UploadScreen --> Submitted: Creator: Submit

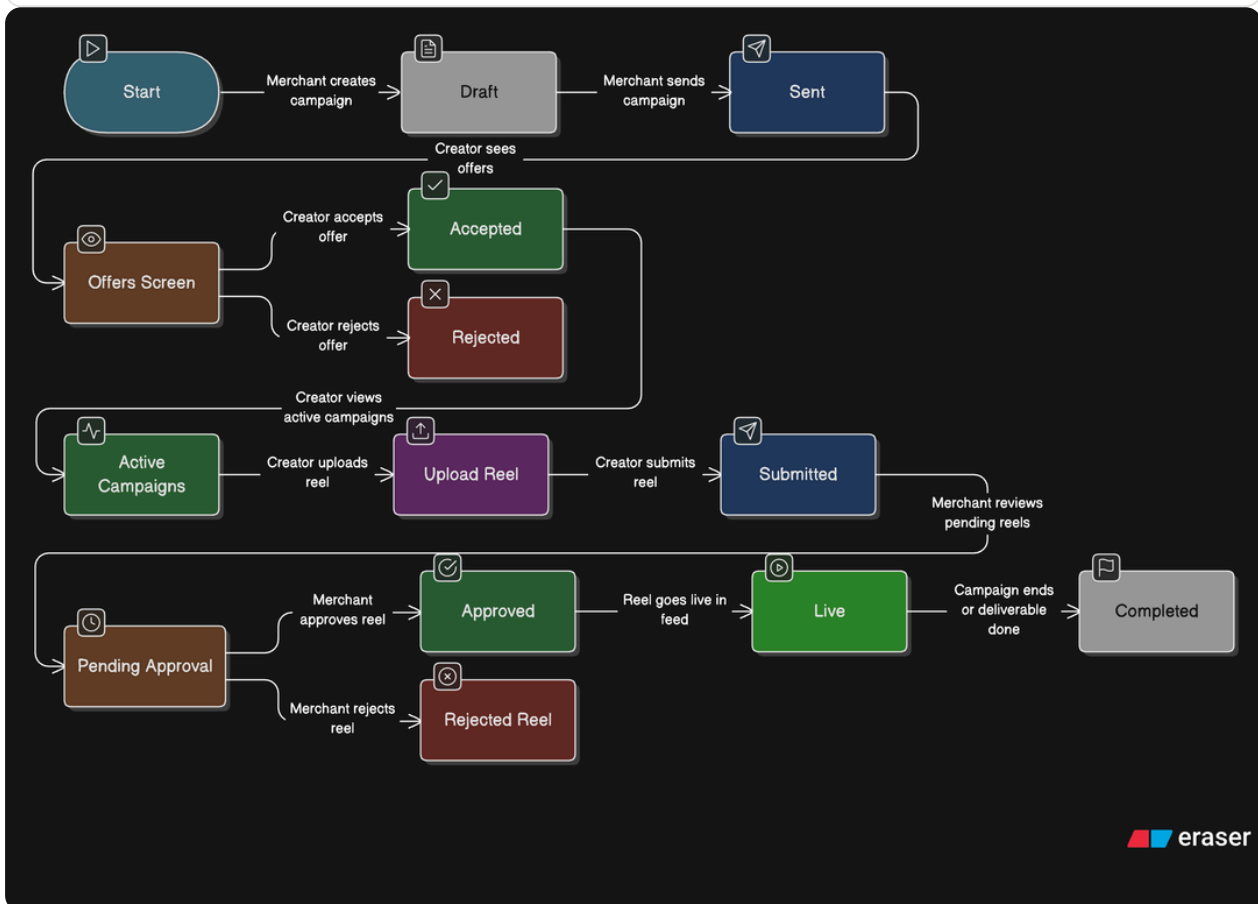
Submitted --> PendingApproval: Merchant: Pending reels

PendingApproval --> Approved: Merchant: Approve

PendingApproval --> RejectedReel: Merchant: Reject

Approved --> Live: Reel live in feed

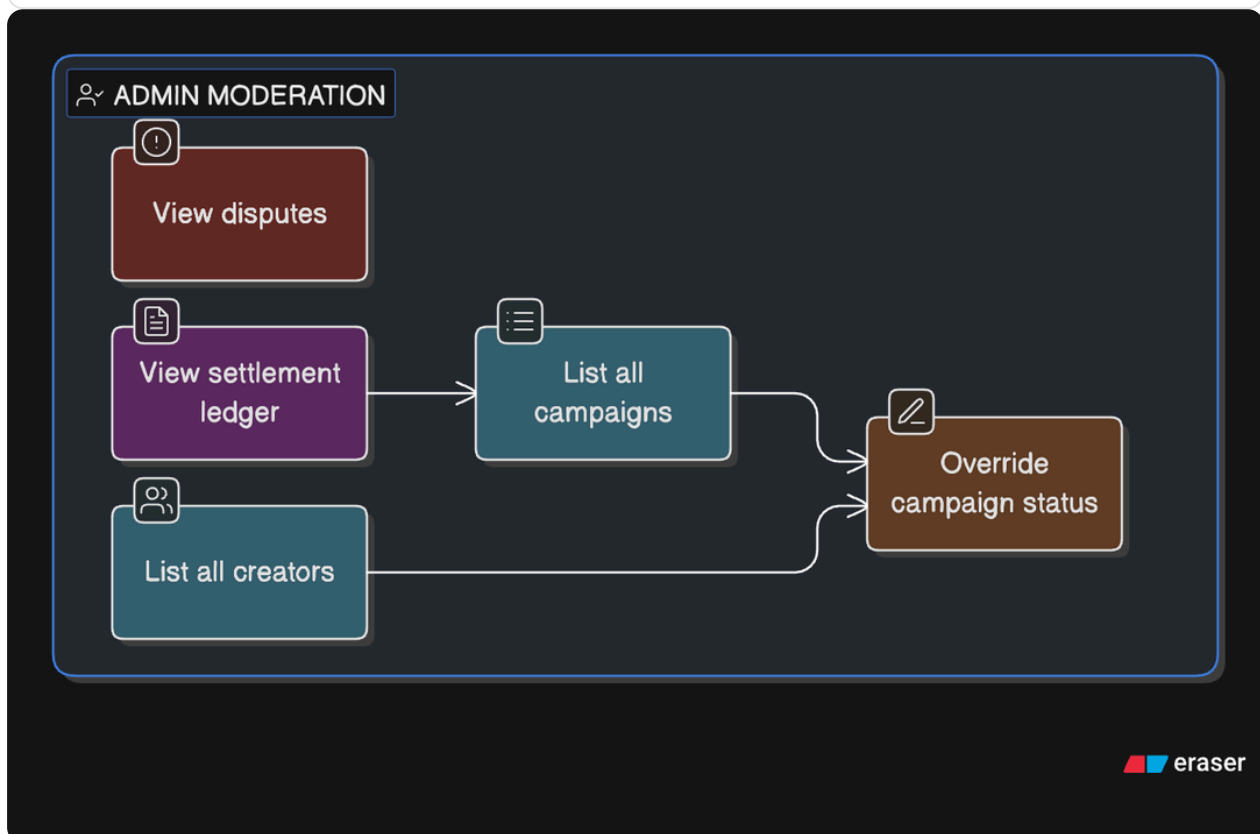
Live --> Completed: Campaign end / deliverable done



6.9 Admin (Optional) — Moderation Section

Mermaid Diagram

```
flowchart TB
    subgraph Admin_Moderation["Admin / Super Admin - Creator Moderation (Optional V1)"]
        CampList[List all campaigns]
        CreatorList[List all creators]
        Override[Override campaign status]
        Disputes[View disputes - V2]
        LedgerView[View settlement ledger]
        CampList --> Override
        CreatorList --> Override
        LedgerView --> CampList
    end
end
```



6.10 Summary: Full Section Map (All Roles)

Mermaid Diagram

flowchart TB

```

subgraph Customer_UI["Customer / User UI"]
    C_ReelFeed[Reel Feed]
    C_Product[Product Page - with attribution click]
    C_Cart[Cart & Checkout]
    C_Orders[My Orders]
end

end

subgraph Creator_UI["Creator UI"]
    Cr_Onboard[Onboarding: Categories, Availability, Portfolio]
    Cr_Home[Home: Stats, deadlines]
    Cr_Categories[My Categories]
    Cr_Portfolio[Portfolio]
    Cr_Deals[Deals: Offers, Active, Completed]
    Cr_Upload[Upload Reel: Select campaign, video, submit]
    Cr_Earnings[Earnings & Payouts]
    Cr_Settings[Settings]
end

end

subgraph Merchant_UI["Merchant UI - New Section"]
    M_Find[Find Creators]
    M_Create[Create Campaign]
    M_Manage[Manage Campaigns: Approve reels, track sales]
    M_Payout[Commission / Payout Summary]
end

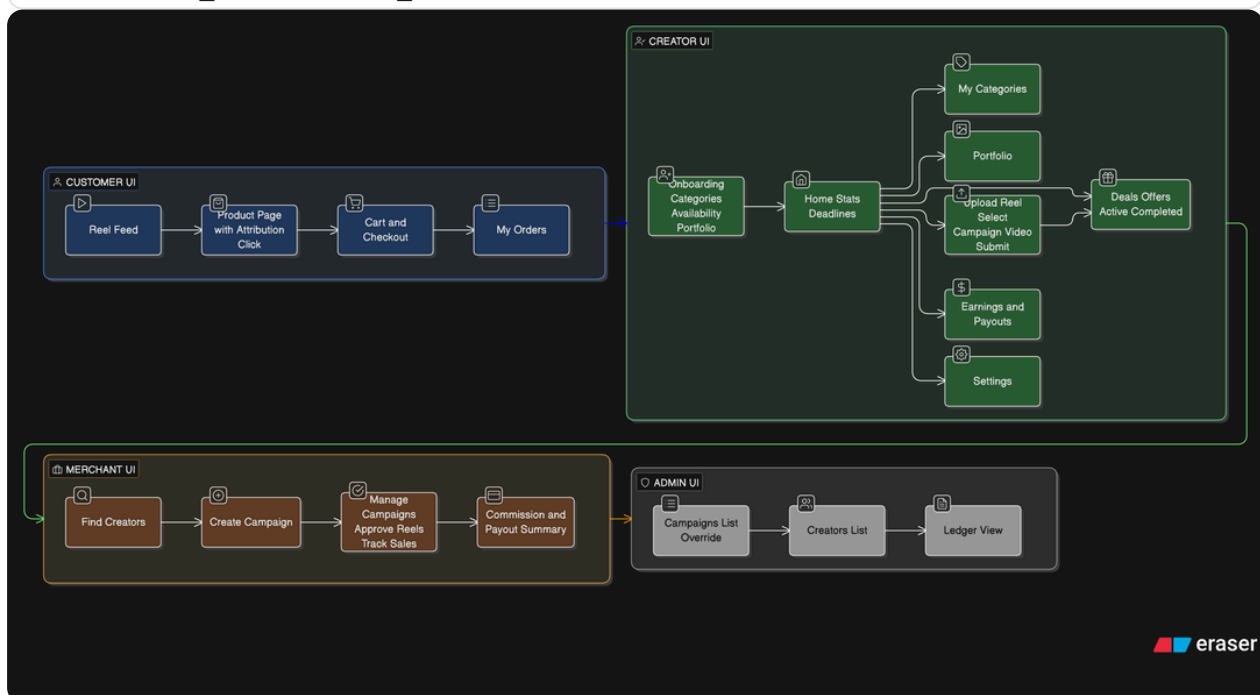
end

subgraph Admin_UI["Admin UI - Optional"]
    A_Campaigns[Campaigns list / override]
    A_Creators[Creators list]
    A_Ledger[Ledger view]
end

end

Customer_UI ~~~ Creator_UI
Creator_UI ~~~ Merchant_UI
Merchant_UI ~~~ Admin_UI

```



6.11 Navigation & Information Architecture (IA)

Role	Primary nav / Sections
Creator	Home, My Categories, Portfolio, Deals, Upload Reel, Earnings, Settings
Merchant	(Existing nav) + Creator Marketing → Find Creators, Create Campaign, Manage Campaigns, Payout Summary
Customer	(Existing) Reel feed, Product page, Cart, Orders — no new top-level section; attribution is invisible (backend click log).
Admin	(Optional) Campaigns, Creators, Ledger / Settlement

7. Data Models

7.1 Current (Relevant) Models

users `id`, `email`, `password_hash`, `first_name`, `last_name`, `phone`, `role` (Enum: user, merchant, admin, super_admin), `is_active`, ...

merchant_profiles `id`, `user_id` (FK users), `business_name`, `business_email`, ... (no creator profile)

reels `reel_id`, `merchant_id` (FK), `product_id` (FK, nullable), `product_url`, `product_name`, `category_id`, `category_name`, `platform`, `video_url`, `description`, `approval_status`, `approved_by`, ...

products `product_id`, `merchant_id`, `category_id`, `brand_id`, ...

categories `category_id`, `parent_id`, `name`, `slug`, ...

orders `order_id`, `user_id`, `order_status`, ...

order_items `order_item_id`, `order_id`, `product_id`, `merchant_id`, quantity, prices, `item_status`, ... (no attribution)

7.2 New / Modified Entities (V1)

User (modify)

- Add to `UserRole` enum: `CREATOR = 'creator'`.

creator_profiles (new)

- `id` (PK)
- `user_id` (FK users, unique) — one creator profile per user
- `availability` — e.g. ENUM or VARCHAR: `available`, `busy`
- `language_preferences` — optional JSON or string
- `portfolio_links` — optional JSON (array of URLs)
- `created_at`, `updated_at`

creator_categories (new)

- `id` (PK)
- `creator_id` (FK creator_profiles)
- `category_id` (FK categories)

- Unique (creator_id, category_id)
- Ensures ≥5 categories per creator (enforced in application).

campaigns (new)

- `campaign_id` (PK)
- `campaign_code` (unique, short, human-friendly)
- `merchant_id` (FK merchant_profiles)
- `creator_id` (FK creator_profiles)
- `product_id` (FK products)
- `status` — ENUM: draft, sent, accepted, active, submitted, approved, live, completed, rejected, cancelled
- `commission_type` — e.g. percent_capped, percent_unlimited
- `commission_percent` — decimal
- `commission_cap_quantity` — nullable int (cap units)
- `campaign_window_start`, `campaign_window_end` — optional
- `deliverable_type` — V1: e.g. "1_reel"
- `created_at`, `updated_at`, `sent_at`, `accepted_at`, `completed_at`

reels (modify)

- Add `campaign_id` (FK campaigns, nullable). Null = merchant-uploaded (current behaviour); non-null = creator-uploaded, campaign-linked.
- For creator reels: `merchant_id` and `product_id` can be derived from campaign; store for convenience/denormalization.
- `approval_status`: for creator reels, support `pending` until merchant approves.

attribution_events (new)

- `id` (PK)
- `user_id` (FK users, nullable for guest)
- `reel_id` (FK reels)
- `campaign_id` (FK campaigns)
- `product_id` (FK products)
- `clicked_at` (datetime)
- Optional: `session_id`, `device_id` for fraud later

order_items (modify)

- Add `attributed_campaign_id` (FK campaigns, nullable). Set when order is placed (or when item becomes eligible) using last-touch rule within attribution window.

settlement_ledger (new)

- `id` (PK)
- `order_item_id` (FK order_items)
- `campaign_id` (FK)
- `creator_id` (FK)

- `merchant_id` (FK)
- `eligible_amount` — net amount (item price – discount – refund) for commission base
- `creator_commission_amount`
- `platform_fee_amount` (5%)
- `merchant_net_amount` (informational)
- `status` — e.g. pending_payout, paid, reversed
- `created_at, paid_at`

creator_payouts (new)

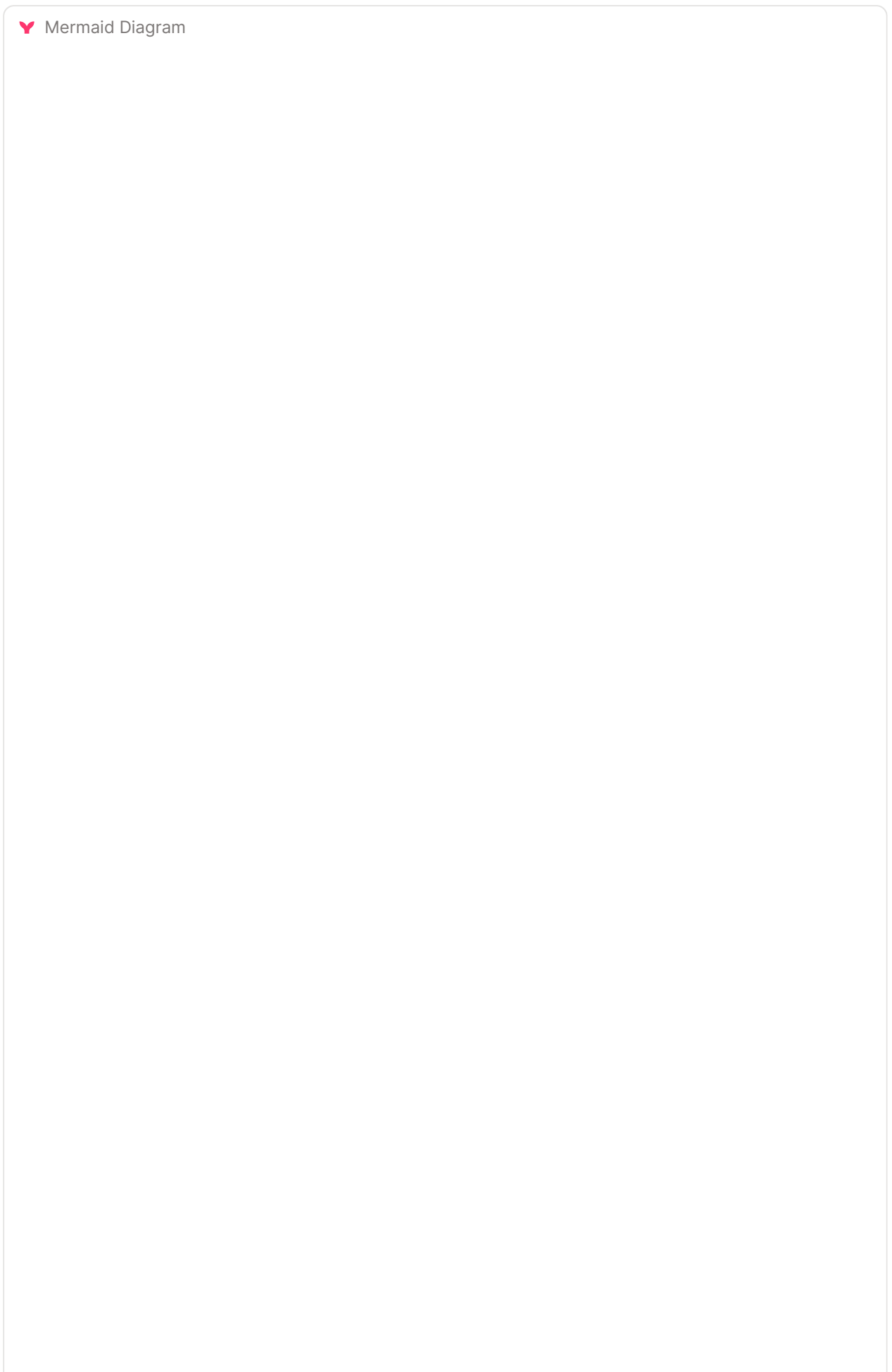
- `id` (PK)
- `creator_id` (FK)
- `amount`
- `status` — e.g. pending, processing, completed, failed
- `period_start, period_end`
- `ledger_entry_ids` or link to ledger rows
- `paid_at, reference` (bank/UPI ref later)

(Exact table names and column types can follow backend conventions; the above is the logical shape.)

7.3 Entity-Relationship & Connectivity

The following diagram shows all Creator-module entities and their relationships. **Solid lines** are direct FKs; **dashed** indicates logical/derived linkage.

Mermaid Diagram



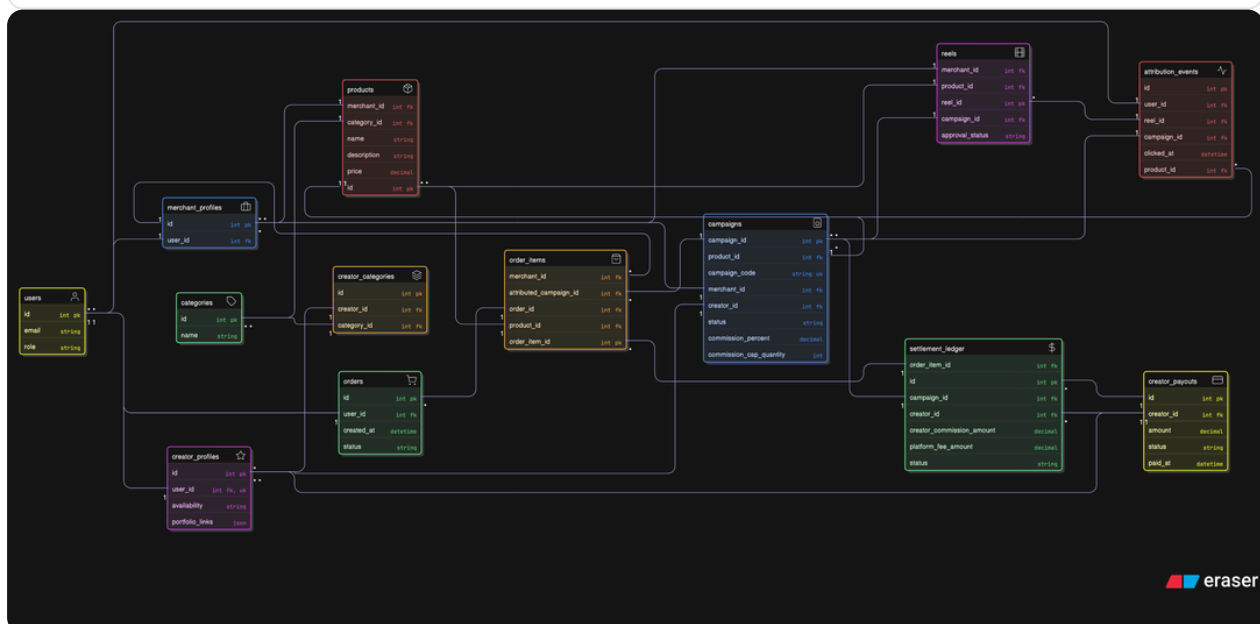
erDiagram

```
USERS ||--o| MERCHANT_PROFILES : "has"
USERS ||--o| CREATOR_PROFILES : "has"
USERS ||--o{ ORDERS : "places"
USERS ||--o{ ATTRIBUTION_EVENTS : "triggers"
CREATOR_PROFILES ||--o{ CREATOR_CATEGORIES : "has"
CREATOR_PROFILES ||--o{ CAMPAIGNS : "assigned_to"
CREATOR_PROFILES ||--o{ CREATOR_PAYOUTS : "receives"
CATEGORIES ||--o{ CREATOR_CATEGORIES : "in"
CATEGORIES ||--o{ PRODUCTS : "contains"
MERCHANT_PROFILES ||--o{ PRODUCTS : "owns"
MERCHANT_PROFILES ||--o{ CAMPAIGNS : "creates"
MERCHANT_PROFILES ||--o{ REELS : "owns_or_approves"
CAMPAIGNS ||--o{ REELS : "linked_to"
CAMPAIGNS }o--|| PRODUCTS : "for_product"
CAMPAIGNS ||--o{ SETTLEMENT_LEDGER : "earns_from"
CAMPAIGNS ||--o{ ATTRIBUTION_EVENTS : "attributed_via"
PRODUCTS ||--o{ REELS : "featured_in"
PRODUCTS ||--o{ ORDER_ITEMS : "sold_as"
PRODUCTS }o--|| CATEGORIES : "belongs_to"
REELS ||--o{ ATTRIBUTION_EVENTS : "click_from"
ORDERS ||--o{ ORDER_ITEMS : "contains"
ORDER_ITEMS }o--o| CAMPAIGNS : "attributed_to"
ORDER_ITEMS ||--o{ SETTLEMENT_LEDGER : "settled_in"
SETTLEMENT_LEDGER }o--o{ CREATOR_PAYOUTS : "included_in"
USERS {
    int id PK
    string email
    string role "user, merchant, creator, admin"
}
CREATOR_PROFILES {
    int id PK
    int user_id FK,UK
    string availability
    json portfolio_links
}
CREATOR_CATEGORIES {
    int id PK
    int creator_id FK
    int category_id FK
}
CAMPAIGNS {
    int campaign_id PK
    string campaign_code UK
    int merchant_id FK
    int creator_id FK
    int product_id FK
    string status
    decimal commission_percent
    int commission_cap_quantity
}
REELS {
    int reel_id PK
    int merchant_id FK
```

```

    int product_id FK
    int campaign_id FK "nullable"
    string approval_status
}
ATTRIBUTION_EVENTS {
    int id PK
    int user_id FK
    int reel_id FK
    int campaign_id FK
    int product_id FK
    datetime clicked_at
}
ORDER_ITEMS {
    int order_item_id PK
    int order_id FK
    int product_id FK
    int merchant_id FK
    int attributed_campaign_id FK "nullable"
}
SETTLEMENT_LEDGER {
    int id PK
    int order_item_id FK
    int campaign_id FK
    int creator_id FK
    decimal creator_commission_amount
    decimal platform_fee_amount
    string status
}
}
CREATOR_PAYOUTS {
    int id PK
    int creator_id FK
    decimal amount
    string status
    datetime paid_at
}
}

```



7.4 Table Connectivity Matrix

From Table	To Table	Relationship	FK Column(s)
creator_profiles	users	N:1	user_id
creator_categories	creator_profiles	N:1	creator_id
creator_categories	categories	N:1	category_id
campaigns	merchant_profiles	N:1	merchant_id
campaigns	creator_profiles	N:1	creator_id
campaigns	products	N:1	product_id
reels	campaigns	N:1 (optional)	campaign_id
reels	merchant_profiles	N:1	merchant_id
reels	products	N:1 (optional)	product_id
attribution_events	users	N:1 (optional)	user_id
attribution_events	reels	N:1	reel_id
attribution_events	campaigns	N:1	campaign_id
attribution_events	products	N:1	product_id
order_items	campaigns	N:1 (optional)	attributed_campaign_id
settlement_ledger	order_items	N:1	order_item_id
settlement_ledger	campaigns	N:1	campaign_id
settlement_ledger	creator_profiles	N:1	creator_id
creator_payouts	creator_profiles	N:1	creator_id

7.5 Data Flow Diagrams (DFD)

7.5.1 DFD Level 0 — Context Diagram

Shows external entities and their high-level data flows with the system.

Mermaid Diagram

flowchart LR

```

subgraph External["External Entities"]
    Creator[Creator]
    Merchant[Merchant]
    Customer[Customer]
    Admin[Admin]
end

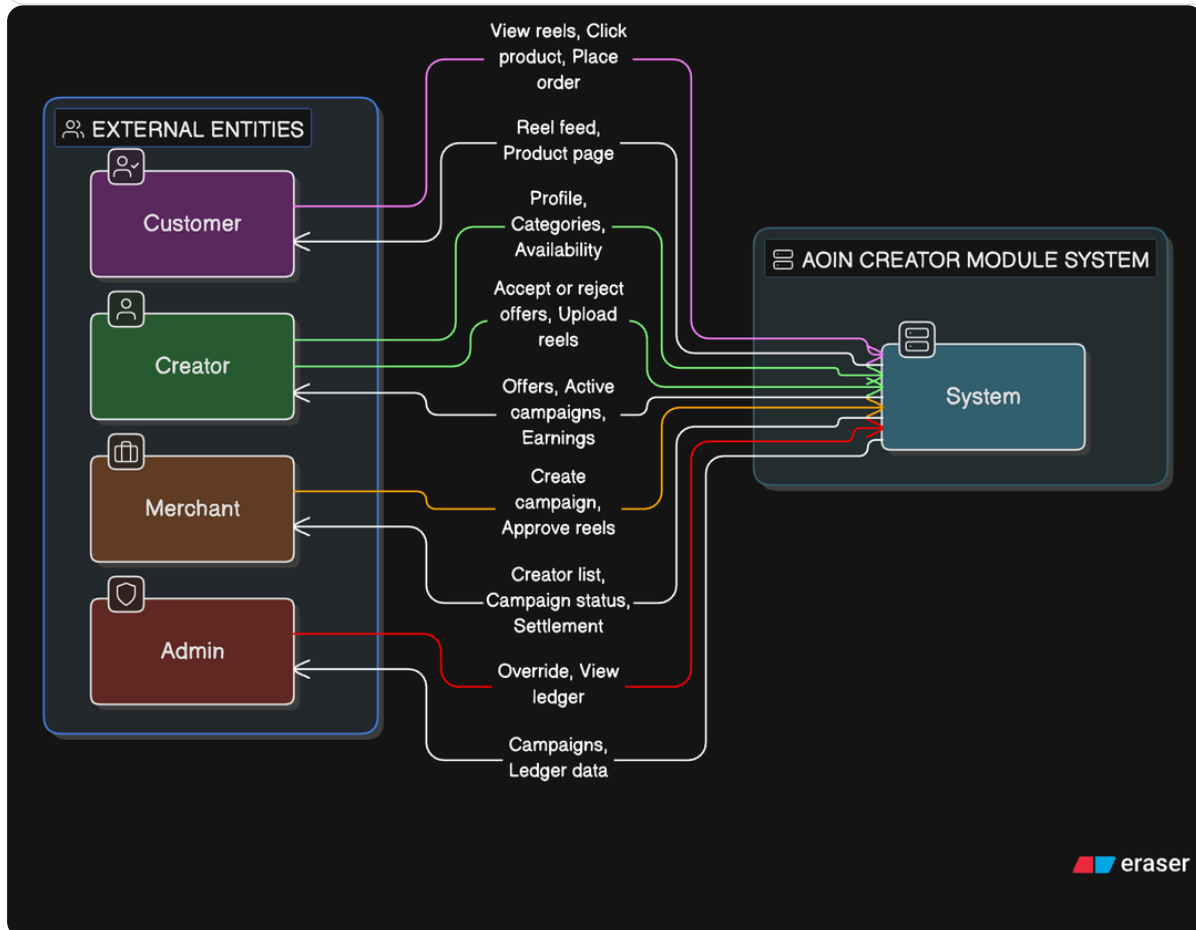
end

subgraph System["AOIN Creator Module System"]
    SYS[(System)]
end

end

Creator -->|Profile, Categories, Availability| SYS
Creator -->|Accept/Reject offers, Upload reels| SYS
SYS -->|Offers, Active campaigns, Earnings| Creator
Merchant -->|Create campaign, Approve reels| SYS
SYS -->|Creator list, Campaign status, Settlement| Merchant
Customer -->|View reels, Click product, Place order| SYS
SYS -->|Reel feed, Product page| Customer
Admin -->|Override, View ledger| SYS
SYS -->|Campaigns, Ledger data| Admin

```



7.5.2 DFD Level 1 — Main Processes & Data Stores

Shows major processes and their inputs/outputs and data stores.

🐙 Mermaid Diagram

flowchart TB

```
    subgraph External["External Entities"]
      Creator[Creator]
      Merchant[Merchant]
      Customer[Customer]
    end

    end

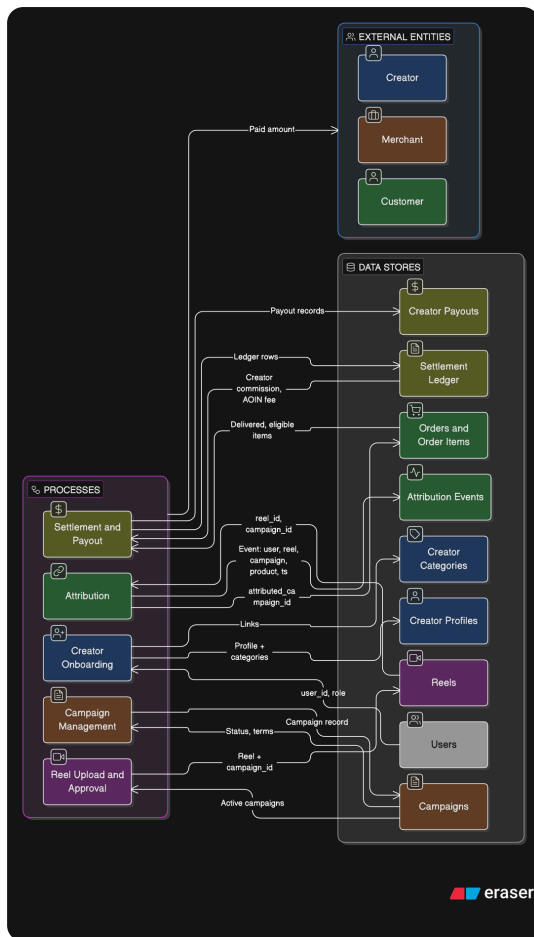
    subgraph Processes["Processes"]
      P1[1. Creator Onboarding]
      P2[2. Campaign Management]
      P3[3. Reel Upload & Approval]
      P4[4. Attribution]
      P5[5. Settlement & Payout]
    end

    end

    subgraph DataStores["Data Stores"]
      D1[(users)]
      D2[(creator_profiles)]
      D3[(creator_categories)]
      D4[(campaigns)]
      D5[(reels)]
      D6[(attribution_events)]
      D7[(orders / order_items)]
      D8[(settlement_ledger)]
      D9[(creator_payouts)]
    end

    end

    Creator -->|Signup, Categories, Availability| P1
    P1 -->|Profile + categories| D2
    P1 -->|Links| D3
    D1 -->|user_id, role| P1
    Merchant -->|Create, Send, Cancel campaign| P2
    Creator -->|Accept, Reject offer| P2
    P2 -->|Campaign record| D4
    D4 -->|Status, terms| P2
    Creator -->|Upload reel - campaign_id, video| P3
    Merchant -->|Approve, Reject reel| P3
    P3 -->|Reel + campaign_id| D5
    D4 -->|Active campaigns| P3
    Customer -->|Click product from reel| P4
    P4 -->|Event: user, reel, campaign, product, ts| D6
    D5 -->|reel_id, campaign_id| P4
    Customer -->|Place order| P4
    P4 -->|attributed_campaign_id| D7
    D7 -->|Delivered, eligible items| P5
    P5 -->|Ledger rows| D8
    P5 -->|Payout records| D9
    D8 -->|Creator commission, AOIN fee| P5
    P5 -->|Paid amount| Creator
```



7.5.3 DFD Level 2 — Attribution Process (Detail)

Detail of how attribution events are recorded and how order items get attributed.

Mermaid Diagram

flowchart LR

```

subgraph Input["Inputs"]
    A[Customer clicks product from reel]
    B[Order placed]
end

end

subgraph AttributionProcess["Attribution Process"]
    P1[Record click event]
    P2[Store: user_id, reel_id, campaign_id, product_id, clicked_at]
    P3[On order: for each item find last touch in window]
    P4[Set order_item.attributed_campaign_id]
    P1 --> P2
    B --> P3
    P3 --> P4
end

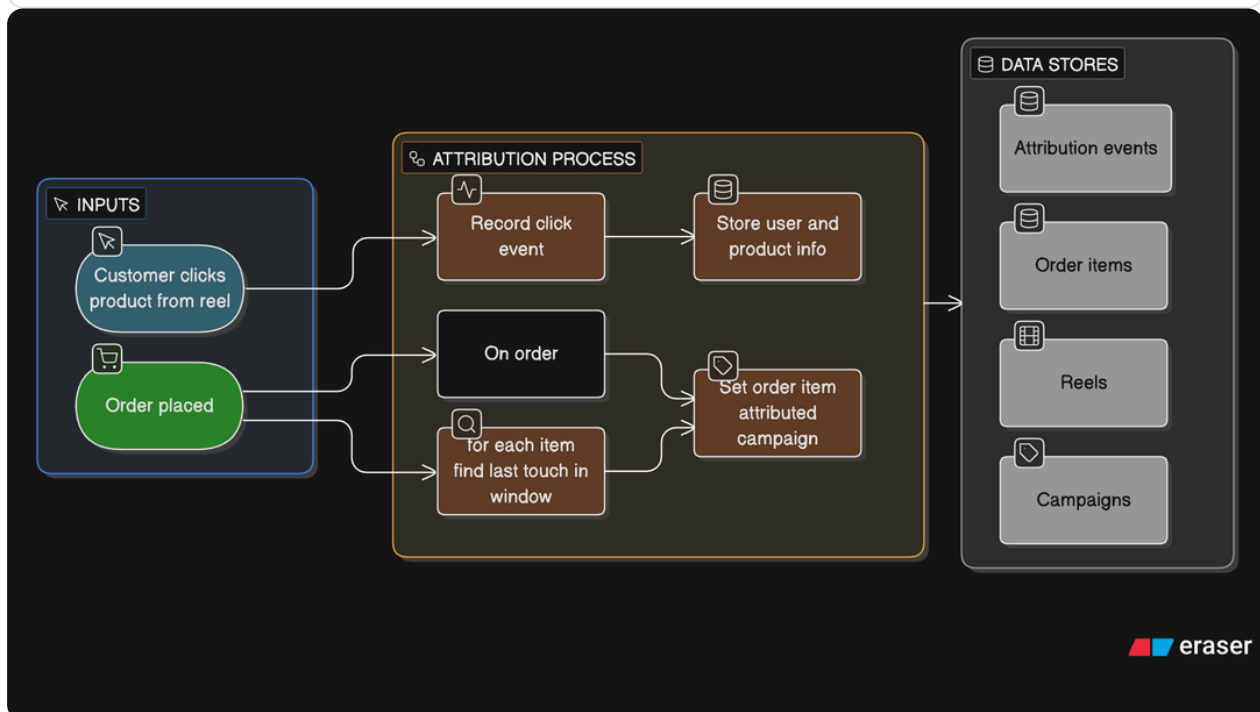
end

subgraph Stores["Data Stores"]
    S1[(attribution_events)]
    S2[(order_items)]
    S3[(reels)]
    S4[(campaigns)]
end

end

A --> P1
P1 --> P2 --> S1
S3 -->|reel_id -> campaign_id| P2
S4 -->|campaign_id| P2
P3 --> S1
S1 -->|last touch within X days| P3
P4 --> S2

```



7.5.4 DFD Level 2 — Settlement Process (Detail)

Detail of how eligible sales become ledger entries and payouts.

Mermaid Diagram

flowchart TB

```

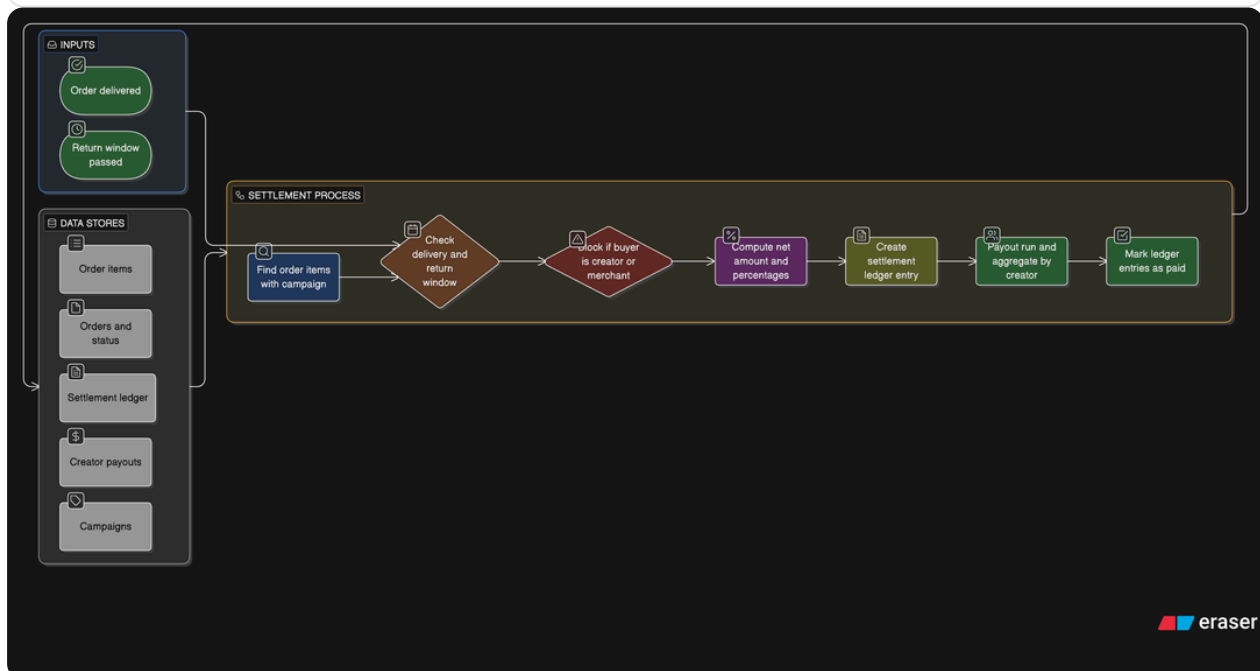
subgraph Input["Inputs"]
    I1[Order delivered]
    I2[Return window passed]
end

subgraph SettlementProcess["Settlement Process"]
    P1[Find order_items with attributed_campaign_id]
    P2[Check delivery + return window]
    P3[Compute net amount, creator %, AOIN 5%]
    P4[Create settlement_ledger row]
    P5[Block if buyer = creator or merchant]
    P6[Payout run: aggregate by creator, create creator_payout]
    P7[Mark ledger entries as paid]
    P1 --> P2 --> P5 --> P3 --> P4
    P4 --> P6 --> P7
end

subgraph Stores["Data Stores"]
    D1[(order_items)]
    D2[(orders / status)]
    D3[(settlement_ledger)]
    D4[(creator_payouts)]
    D5[(campaigns)]
end

I1 --> P2
I2 --> P2
D1 --> P1
D2 --> P2
D5 --> P3
P4 --> D3
P6 --> D4
D3 --> P6
P7 --> D3

```



7.5.5 Database Data Flow Summary

Endpoint	200	201	400	401	403	404	409	413
POST /api/auth/signup		✓	✓					
POST /api/auth/login	✓		✓	✓				
POST /api/creator/onboarding		✓	✓	✓	✓		✓	
GET /api/creator/profile	✓			✓	✓	✓		
PATCH /api/creator/profile	✓		✓	✓	✓	✓		
GET /api/categories	✓							
GET /api/campaigns/creators	✓			✓	✓			
POST /api/campaigns		✓	✓		✓			
POST /api/campaigns/:id/send	✓		✓	✓	✓	✓		
GET /api/campaigns	✓			✓	✓			
GET /api/campaigns/:id	✓			✓	✓	✓		
PATCH /api/campaigns/:id	✓		✓	✓	✓	✓		
GET /api/creator/campaigns	✓			✓	✓			
GET /api/creator/campaigns/:id	✓			✓	✓	✓		
POST /api/creator/campaigns/:id/accept	✓		✓	✓	✓	✓		
POST /api/creator/campaigns/:id/reject	✓			✓	✓	✓		
GET /api/creator/campaigns/active	✓			✓	✓			
POST /api/creator/reels		✓	✓	✓	✓			✓

GET /api/campaigns/:id /reels/pending	✓			✓	✓	✓		
POST /api/reels/:id/approve	✓		✓	✓	✓	✓		
POST /api/reels/:id/reject	✓		✓	✓	✓	✓		
POST /api/attribution/click		✓	✓	✓				
GET /api/creator/earnings/summary	✓			✓	✓			
GET /api/creator/earnings	✓			✓	✓			
GET /api/creator/payouts/:id	✓			✓	✓	✓		
GET /api/merchant/campaigns/:id/settlement	✓			✓	✓	✓		

8.1 Auth & Creator Signup

8.1.1 Signup (with role creator)

Purpose: Register a new user with role `creator`. Same as existing signup; request body includes `role: "creator"`.

Field	Type	Location	Required	Description
email	string	body	Yes	Valid email.
password	string	body	Yes	Min length per policy.
first_name	string	body	Yes	
last_name	string	body	Yes	
phone	string	body	No	
role	string	body	Yes	Must be <code>"creator"</code> for creator signup.

Request (example):


```
{  
  "email": "creator@example.com",  
  "password": "SecurePass123",  
  "first_name": "Jane",  
  "last_name": "Creator",  
  "phone": "+919876543210",  
  "role": "creator"  
}
```

Response — 201 Created:

```
{  
  "message": "User registered successfully",  
  "user": {  
    "id": 42,  
    "email": "creator@example.com",  
    "first_name": "Jane",  
    "last_name": "Creator",  
    "role": "creator",  
    "is_email_verified": false  
  },  
  "access_token": "eyJ...",  
  "refresh_token": "eyJ...",  
  "expires_in": 3600  
}
```

Response — 400 Bad Request (e.g. validation):

```
{  
  "error": "Validation failed",  
  "code": "VALIDATION_ERROR",  
  "details": { "field": "email", "message": "Email already registered" }  
}
```

8.1.2 Login (creator)

Purpose: Same as existing login. Response includes `role`; frontend redirects creators to Creator Dashboard.

Request: Existing login body (email/phone + password).

Response — 200 OK: Same as existing (tokens + user with `role: "creator"`).

8.2–8.11 Full request/response for every API

For each and every endpoint (onboarding, profile, discover creators, create/send/cancel campaign, list/get campaign, accept/reject, active campaigns, upload reel, pending reels, approve/reject reel, attribution click, earnings summary/list, payout detail, campaign settlement, admin APIs) **with expected request and response bodies**, see:

CREATOR_MODULE_API_SPEC.md (./CREATOR_MODULE_API_SPEC.md)

That document lists all APIs in order with:

- Method, path, auth
- Request body/query parameters and example JSON
- Response 200/201 success body
- Response 400/404/409 (and 413 where applicable) error body

8.12 API Quick Reference Table

#	Method	Path	Auth	Purpose
8.1.1	POST	/api/auth/signup	—	Signup with role creator
8.1.2	POST	/api/auth/login	—	Login (creator redirect)
8.2.1	POST	/api/creator/onboarding	Creator	Complete onboarding
8.2.2	GET	/api/creator/profile	Creator	Get profile + categories
8.2.3	PATCH	/api/creator/profile	Creator	Update profile
8.2.4	GET	/api/categories	Optional	List categories
8.3.1	GET	/api/campaigns/creators	Merchant	Discover creators
8.3.2	POST	/api/campaigns	Merchant	Create campaign (draft)
8.3.3	POST	/api/campaigns/:id/send	Merchant	Send offer
8.3.4	GET	/api/campaigns	Merchant	List my campaigns
8.3.5	GET	/api/campaigns/:id	Merchant	Get campaign detail
8.3.6	PATCH	/api/campaigns/:id	Merchant	Cancel / Extend
8.4.1	GET	/api/creator/campaigns	Creator	List my campaigns
8.4.2	GET	/api/creator/campaigns/:id	Creator	Get my campaign
8.4.3	POST	/api/creator/campaigns/:id/accept	Creator	Accept offer

		t		
8.4.4	POST	/api/creator/campaigns/:id/reject	Creator	Reject offer
8.4.5	GET	/api/creator/campaigns/active	Creator	Active campaigns (dropdown)
8.5.1	POST	/api/creator/reels	Creator	Upload reel (campaign-linked)
8.6.1	GET	/api/campaigns/:id/reels/pending	Merchant	Pending reels for campaign
8.6.2	POST	/api/reels/:id/approve	Merchant	Approve reel
8.6.3	POST	/api/reels/:id/reject	Merchant	Reject reel
8.7.1	POST	/api/attribution/click	User/guest	Record reel→product click
8.8.1	GET	/api/creator/earnings/summary	Creator	Earnings summary
8.8.2	GET	/api/creator/earnings	Creator	List earnings
8.8.3	GET	/api/creator/payouts/:id	Creator	Payout detail
8.9.1	GET	/api/merchant/campaigns/:id/settlement	Merchant	Campaign settlement

Legacy summary (pre-expansion):

- **Auth:** Signup, login, JWT, role in token. No creator signup/onboarding.
- **Reels:**
 - Upload (merchant only), list (public, merchant, by product), view, like, share, delete, etc.
 - No campaign-scoped upload, no “my campaigns” or “pending approval”.
- **Products/Categories:** CRUD for merchants; public read. No “creators by category”.
- **Orders:** Create, list, status, shipments. No attribution or settlement APIs.

8.2 New / Modified APIs (V1)

Auth & Creator onboarding

- Signup with `role=creator` (or convert to creator).
- `POST /api/creator/onboarding` — set categories (≥5), availability, portfolio (optional).
- `GET/PATCH /api/creator/profile` — get/update creator profile and categories.

Campaigns (Merchant)

- `GET /api/campaigns/creators` — discover creators (filter by category, availability).
- `POST /api/campaigns` — create campaign (product_id, creator_id, terms).
- `POST /api/campaigns/:id/send` — send offer (Draft → Sent).
- `GET /api/campaigns` — list merchant's campaigns (filter by status).
- `PATCH /api/campaigns/:id` — cancel, extend window, etc.
- `GET /api/campaigns/:id` — campaign detail (terms, status, linked reels).

Campaigns (Creator)

- `GET /api/creator/campaigns` — my offers (sent), active, completed.
- `POST /api/creator/campaigns/:id/accept` — accept offer.
- `POST /api/creator/campaigns/:id/reject` — reject offer.

Reels (Creator path)

- `GET /api/creator/campaigns/active` — list active campaigns (for “Select Campaign” dropdown).
- `POST /api/creator/reels` — upload reel (campaign_id, video, description). Validates campaign is active and assigned to creator; sets approval_status = pending.

Reels (Merchant approval)

- `GET /api/campaigns/:id/reels/pending` — reels pending approval for this campaign.
- `POST /api/reels/:id/approve` — approve reel (creator reel only).
- `POST /api/reels/:id/reject` — reject with reason.

Attribution

- `POST /api/attribution/click` — record reel→product click (user from JWT or session, reel_id, product_id; backend resolves campaign_id). Called by frontend when user taps product from reel.

Settlement & Earnings

- `GET /api/creator/earnings` — pending, eligible, paid (summary and list).
- `GET /api/merchant/campaigns/:id/settlement` — attributed sales, commission breakdown.
- (Internal or admin) Process eligibility (e.g. cron after delivery + return window), create ledger rows, run payouts.

Reels (read) — optional changes

- Include `campaign_id` and creator info in reel response where applicable; keep backward compatibility for merchant-only reels.

9. Implementation Phases & Deliverables

Phase 1: Foundation (Roles & Creator Profile)

- Add `UserRole.CREATOR`.
- Create `creator_profiles` and `creator_categories` tables; migrations.
- Creator signup/onboarding API: set role, create profile, set ≥5 categories, availability.

- `GET/PATCH /api/creator/profile`.
- **Deliverable:** Creators can sign up and complete onboarding.

Phase 2: Campaign / Deal Module

- Create `campaigns` table; migration.
- Merchant: discover creators (by category, availability), create campaign, send offer.
- Creator: list offers, accept/reject.
- Status transitions (Draft → Sent → Accepted → Active; Rejected, Cancelled).
- **Deliverable:** Merchants can create and send offers; creators can accept/reject; `campaign_code` generated.

Phase 3: Creator Reel Upload & Merchant Approval

- Add `campaign_id` (nullable) to `reels`; migration.
- Creator upload API: select campaign (active, assigned), upload video + description; create reel with `campaign_id`, `approval_status` = pending.
- Merchant approval APIs: list pending reels for campaign, approve/reject.
- Visibility: creator reels appear in feed only when approved.
- **Deliverable:** Creators upload campaign-linked reels; merchants approve; reels go live after approval.

Phase 4: Attribution

- Create `attribution_events` table; migration.
- `POST /api/attribution/click` — store event (user, reel, product, campaign, timestamp).
- On order placement (or when item is created): for each item, resolve last-touch within X days (e.g. 7); set `attributed_campaign_id` on `order_item` (or equivalent).
- **Deliverable:** Reel→product clicks logged; orders/items linked to campaign for eligible purchases.

Phase 5: Settlement & Payouts

- Add `attributed_campaign_id` (or equivalent) to `order_items`; create `settlement_ledger` and `creator_payouts`; migrations.
- Eligibility job: after delivery + return window, create ledger entries (creator commission, AOIN fee, merchant net).
- Block attribution if buyer `user_id` == creator or merchant (same user).
- Creator earnings API: pending, eligible, paid.
- Merchant campaign settlement API: attributed sales, commission summary.
- Payout run (weekly/monthly or threshold): mark ledger as paid, create `creator_payout` record (V1 can be manual trigger).
- Refund/return: reversal logic for creator and AOIN fee.

- **Deliverable:** Commission and platform fee calculated; creator earnings visible; payouts runnable (manual or scheduled).

Phase 6: Polish & Edge Cases

- Reel upload: one reel → one product (campaign enforces product).
- Fraud: block attribution when buyer is creator or merchant.
- Notifications: new offer, accepted, reel submitted, reel approved/rejected (optional).
- Admin: list campaigns, override status, view ledger (optional).

10. Edge Cases & Business Rules

Case	Rule
Creator categories	Minimum 5 categories mandatory at onboarding; edit allowed with same minimum.
Portfolio	Optional but recommended; no hard block.
Availability	Toggle Available/Busy; used in creator discovery.
Wrong reel / quality	Merchant can reject or request changes; reel not live until approved.
Self-order / fraud (V1)	Do not attribute if order's user_id equals creator's user_id or merchant's user_id.
Multiple products in one reel	V1: one reel → one product (enforced by campaign).
Commission base	Net = item price – item discount – refunds; exclude shipping/tax.
AOIN fee	5% on attributed sales only.
Attribution window	Configurable X days (e.g. 7); last-touch wins.
Eligibility	Order delivered + return/refund window passed (e.g. 7/14 days).
Refund/return	Reverse or adjust creator payout and AOIN fee for affected items.

11. Glossary

Term	Definition
Campaign	A deal between merchant and creator for a specific product: commission terms, optional cap, window; deliverable (e.g. 1 reel).
Campaign code	Short, human-friendly identifier for a campaign (e.g. for dropdown or sharing).

Creator	User with role Creator; has creator profile (categories, portfolio, availability); accepts campaigns and uploads campaign-linked reels.
Attribution	Assigning a sale (order item) to a reel/campaign/creator when the user clicked from that reel to product within the attribution window (last-touch).
Last-touch	Attribution rule: the most recent reel→product click within X days before purchase gets credit.
Eligible sale	Order item that is delivered and past return/refund window, used for settlement.
Settlement ledger	Record of creator commission, AOIN fee, and merchant net per eligible attributed sale.
Creator payout	Payment to creator (batch or single) for earned commission (V1: manual or scheduled run).

12. Architecture Gaps & Missing Specifications

As a Software Architect, the following items are **missing or underspecified** in this PRD. They should be decided and documented before or during implementation to avoid rework and production issues.

12.1 Non-Functional Requirements (NFRs)

Gap	Description	Recommendation
Performance	No latency or throughput targets.	Define: API p95 latency (e.g. < 500ms for read, < 2s for upload), attribution write throughput, reel upload max duration.
Scalability	No expected scale.	Define: max creators, campaigns per merchant, attribution events/day; plan for read-heavy (feed) vs write-heavy (clicks).
Availability	No SLA.	Define: target uptime for Creator APIs; dependency on order/delivery status for settlement.
Rate limiting	Not mentioned.	Define: per-user/per-role limits (e.g. attribution/click, campaign create, reel upload) to prevent abuse.

12.2 Security & Authorization

Gap	Description	Recommendation
RBAC matrix	Who can call what is implied but not tabular.	Add: matrix (Creator / Merchant / Admin × resource × action); e.g. Creator can only PATCH own profile, only accept/reject campaigns assigned to them.

Resource-level auth	Campaign/reel ownership checks are implied.	Specify: merchant can only see/edit own campaigns; creator only campaigns where creator_id = self; admin override rules.
PII & sensitive data	Creator earnings, merchant settlement.	Define: who can see what; audit log for payout, approval, cancel; mask in logs.
JWT / claims	Role in token mentioned; scope not.	Define: token claims (role, creator_id/merchant_id when applicable); token refresh for long sessions.

12.3 Configuration & Constants

Gap	Description	Recommendation
Attribution window	"X days (e.g. 7)" not bound.	Define: config key (e.g. <code>ATTRIBUTION_WINDOW_DAYS</code>), default 7, env or DB.
AOIN fee %	"5%" hardcoded in narrative.	Define: config key (e.g. <code>PLATFORM_FEE_PERCENT</code>), default 5.
Eligibility return window	"7/14 days" — source of truth?	Tie to existing order/shipment config; document which status + which window.
campaign_code format	"Short, human-friendly" only.	Define: format (e.g. <code>CAMP-{alnum}</code>), length, uniqueness, generation (timestamp + random?).
Reel upload limits	Max file size, formats.	Document: max size (e.g. 100MB), allowed types (mp4, webm, mov); align with existing reel upload.

12.4 Integration & System Boundaries

Gap	Description	Recommendation
When attribution is set	"On order placement or when item becomes eligible" is ambiguous.	Decide: set <code>attributed_campaign_id</code> at order placement (synchronous) vs at delivery/eligibility (async job). Document and reflect in DFD/sequence.
Order/delivery source	Eligibility depends on "delivered" and "return window passed".	Document: which order/shipment statuses and fields drive eligibility; link to existing enums and workflows.
Payout mechanism	"Bank/UPI later" vague.	Specify: V1 manual transfer vs integrated payout (Razorpay, PayU, etc.); creator bank/UPI capture and validation (KYC if needed).
Reel storage	Creator reels same as merchant reels?	Confirm: same S3/Cloudinary bucket and path policy; any quota per creator/campaign.
Notifications	Optional; not specified.	

		Decide: in-app only vs email/push; events (new offer, accepted, reel submitted, approved/rejected); payload and channels.
--	--	---

12.5 Data & Compliance

Gap	Description	Recommendation
Retention	attribution_events, ledger, payouts.	Define: retention (e.g. attribution_events 1 year, ledger 7 years); archival or purge policy.
GDPR / deletion	Creator deletes account.	Define: cascade (anonymize/delete profile, categories, campaigns as creator? reels? ledger?); legal hold for payouts.
Currency	Ledger and payouts.	Define: single currency (e.g. INR) or multi-currency; store currency per ledger row and payout.
Data residency	Not mentioned.	If required, state where creator/campaign/attribution data must reside.

12.6 Concurrency & Consistency

Gap	Description	Recommendation
Idempotency	Accept/reject, create campaign, upload reel.	Define: idempotency key (header or body) for POST; behaviour on duplicate (return 200 + existing resource).
Campaign state transitions	State machine drawn; allowed transitions per role not tabular.	Add: matrix (current_status × action × role → new_status); reject invalid transitions with 409.
Double accept	Creator clicks Accept twice.	Idempotent by campaign_id + creator; return 200 with current state if already accepted.
Optimistic locking	Concurrent update to campaign/reel.	Consider version or updated_at check on PATCH to avoid lost updates.
Attribution race	Order placed same second as click.	Define: ordering (e.g. click before order timestamp); or accept both and use last touch.

12.7 Attribution Details

Gap	Description	Recommendation
Guest users	user_id null; "session_id or device_id" optional.	Define: how guest last-touch is keyed (session_id, device_id, cookie); retention and matching at order.
Multi-item order	Each order_item can have different product.	

		Confirm: each order_item gets its own last-touch per product_id; document in attribution rule.
Product mismatch	User clicks product A from reel, adds product B (variant or wrong product).	Define: attribute by product_id match only; or by product + variant/category.

12.8 Settlement & Payout

Gap	Description	Recommendation
Eligibility job	"After delivery + return window".	Specify: trigger (cron schedule vs event); idempotent (skip already-settled items); failure alert and retry.
Payout run	"Weekly/monthly or threshold".	Specify: schedule or threshold; who triggers (admin vs automated); output (CSV for bank, or API to payment provider).
Partial refund	One item in order refunded.	Define: proportional reversal of creator_commission and platform_fee for that item; ledger reversal entry.
Creator onboarding for payout	Bank/UPI capture.	Define: when required (before first payout?); validation; link to existing KYC if any.

12.9 Data Model Gaps

Gap	Description	Recommendation
reels.creator_id	Reel uploaded by creator; ownership is merchant (campaign).	Decide: add optional <code>creator_id</code> on reels for "uploaded by" display and filtering; or always derive from campaign.
Ledger/payout currency	settlement_ledger, creator_payouts.	Add: currency column (e.g. INR); or document single-currency assumption.
creator_payouts	payment_method, beneficiary details.	Add: payment_method (bank/UPI), reference_id, beneficiary_id or mask; for audit and reconciliation.
campaign_code uniqueness	Unique per tenant/global?	Specify: global unique; format and collision handling.

12.10 Observability

Gap	Description	Recommendation
Logging	What to log.	Define: structured logs; PII redaction (email, name in logs); audit events (campaign sent, accepted, reel approved, payout run).
Metrics	Not mentioned.	

		Define: counters (campaigns_created, reels_uploaded, attribution_events, payouts_completed); gauges (pending_reels, eligible_earnings); latency histograms per API.
Tracing	Cross-service.	If order/settlement are separate: add trace id propagation for attribution and eligibility.
Alerting	Not mentioned.	Define: alerts (eligibility job failure, payout run failure, spike in attribution or errors).

12.11 Deployment & Rollout

Gap	Description	Recommendation
Feature flag	Roll out by segment.	Define: flag (e.g. creator_module_enabled, or per-merchant); default off until stable.
Backward compatibility	Existing reels without campaign_id.	Document: all existing reels remain; new fields nullable; feed and APIs handle both.
Migration order	DB and app.	Document: run reels migration (e.g. 006) before deploying creator APIs; order of new tables (creator_profiles → campaigns → attribution_events → settlement_ledger).
API versioning	/api/... vs /v1/...	Decide: prefix (e.g. /v1/creator/...) or same /api with backward-compatible changes.

12.12 Testing

Gap	Description	Recommendation
Test strategy	Not mentioned.	Define: unit (services, state machine), integration (API + DB), e2e (create campaign → accept → upload → approve → order → settlement); contract tests for public APIs.
Load / stress	Not mentioned.	Define: target load for attribution click and reel upload; basic load test before launch.

12.13 Role & Identity

Gap	Description	Recommendation
Dual role	Can one user be both merchant and creator?	State explicitly: V1 one user = one role (no dual); or allow and define behaviour (separate creator_profile and merchant_profile).
Same email	User signs up as creator then as merchant (or vice versa).	Define: same account two roles vs two accounts; login and dashboard routing by role.

12.14 Campaign & Reel Lifecycle Edge Cases

Gap	Description	Recommendation
Campaign window expired	Window end passed; campaign still "active"?	Define: automatic transition to completed/expired when <code>window_end < now</code> ; creator cannot upload after window end.
Reel submitted before window end, approved after	Allowed?	Define: allow; approval can be after window; reel stays linked to campaign for attribution.
Product deleted or out of stock	Campaign references product.	Define: campaign remains; reel visibility follows existing product rules; no new attribution for that product.
Creator deactivated	User disabled or creator profile removed.	Define: existing campaigns/reels/ledger preserved; no new offers to that creator; payouts still processed.

12.15 Documentation & Process

Gap	Description	Recommendation
Changelog	API and behaviour changes.	Maintain: API changelog (new fields, new endpoints, breaking changes) and link from PRD.
Attribution resolution sequence	Synchronous vs async.	Add: sequence diagram for "order placed → resolve last touch → set attributed_campaign_id" once decided.
Ownership & approval	PRD owner, review cycle.	Define: who owns PRD updates; review before each phase; version in Document History.

13. Implementation Reference

This section provides **implementation-ready** artifacts: DDL, state machine table, RBAC matrix, config spec, error codes, sequence diagrams, test scenarios, and rollback/failure handling. Use it as the single reference during build.

13.1 DDL / Migration Snippets (MySQL)

Run in order. Replace `ecommerce_db` with your database name if different.

13.1.1 Add CREATOR to UserRole (application enum)In code: add `CREATOR = 'creator'` to `UserRole` enum in `auth/models/models.py`. No DB change if role is stored as string/VARCHAR.

13.1.2 Create creator_profiles

SQL

```
CREATE TABLE IF NOT EXISTS creator_profiles (  
  id INT NOT NULL AUTO_INCREMENT,  
  user_id INT NOT NULL,  
  availability VARCHAR(20) NOT NULL DEFAULT 'available',  
  language_preferences VARCHAR(100) NULL,  
  portfolio_links JSON NULL,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  UNIQUE KEY uk_creator_profiles_user_id (user_id),  
  CONSTRAINT fk_creator_profiles_user FOREIGN KEY (user_id) REFERENCES users(id) ON  
DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.3 Create creator_categories

SQL

```
CREATE TABLE IF NOT EXISTS creator_categories (  
  id INT NOT NULL AUTO_INCREMENT,  
  creator_id INT NOT NULL,  
  category_id INT NOT NULL,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  UNIQUE KEY uk_creator_categories_creator_category (creator_id, category_id),  
  CONSTRAINT fk_creator_categories_creator FOREIGN KEY (creator_id) REFERENCES  
creator_profiles(id) ON DELETE CASCADE,  
  CONSTRAINT fk_creator_categories_category FOREIGN KEY (category_id) REFERENCES  
categories(category_id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.4 Create campaigns

SQL

```
CREATE TABLE IF NOT EXISTS campaigns (  
    campaign_id INT NOT NULL AUTO_INCREMENT,  
    campaign_code VARCHAR(32) NOT NULL,  
    merchant_id INT NOT NULL,  
    creator_id INT NOT NULL,  
    product_id INT NOT NULL,  
    status VARCHAR(32) NOT NULL DEFAULT 'draft',  
    commission_type VARCHAR(32) NOT NULL,  
    commission_percent DECIMAL(5,2) NOT NULL,  
    commission_cap_quantity INT NULL,  
    campaign_window_start DATETIME NULL,  
    campaign_window_end DATETIME NULL,  
    deliverable_type VARCHAR(32) NOT NULL DEFAULT '1_reel',  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
    sent_at DATETIME NULL,  
    accepted_at DATETIME NULL,  
    completed_at DATETIME NULL,  
    PRIMARY KEY (campaign_id),  
    UNIQUE KEY uk_campaigns_code (campaign_code),  
    KEY idx_campaigns_merchant (merchant_id),  
    KEY idx_campaigns_creator (creator_id),  
    KEY idx_campaigns_status (status),  
    CONSTRAINT fk_campaigns_merchant FOREIGN KEY (merchant_id) REFERENCES  
merchant_profiles(id) ON DELETE CASCADE,  
    CONSTRAINT fk_campaigns_creator FOREIGN KEY (creator_id) REFERENCES  
creator_profiles(id) ON DELETE CASCADE,  
    CONSTRAINT fk_campaigns_product FOREIGN KEY (product_id) REFERENCES  
products(product_id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.5 Alter reels — add campaign_id and creator_id*Prerequisite: migration 006 (product_id nullable, external columns) must be applied first.*

SQL

```
ALTER TABLE reels ADD COLUMN campaign_id INT NULL AFTER product_id;  
ALTER TABLE reels ADD COLUMN creator_id INT NULL AFTER campaign_id;  
ALTER TABLE reels ADD KEY idx_reels_campaign (campaign_id);  
ALTER TABLE reels ADD KEY idx_reels_creator (creator_id);  
-- Add FKs after campaigns table exists:  
ALTER TABLE reels ADD CONSTRAINT fk_reels_campaign FOREIGN KEY (campaign_id)  
REFERENCES campaigns(campaign_id) ON DELETE SET NULL;  
ALTER TABLE reels ADD CONSTRAINT fk_reels_creator FOREIGN KEY (creator_id) REFERENCES  
creator_profiles(id) ON DELETE SET NULL;
```

13.1.6 Create attribution_events

SQL

```
CREATE TABLE IF NOT EXISTS attribution_events (  
  id BIGINT NOT NULL AUTO_INCREMENT,  
  user_id INT NULL,  
  reel_id INT NOT NULL,  
  campaign_id INT NOT NULL,  
  product_id INT NOT NULL,  
  clicked_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  session_id VARCHAR(255) NULL,  
  PRIMARY KEY (id),  
  KEY idx_attribution_user_clicked (user_id, product_id, clicked_at),  
  KEY idx_attribution_campaign (campaign_id),  
  CONSTRAINT fk_attribution_user FOREIGN KEY (user_id) REFERENCES users(id) ON  
DELETE SET NULL,  
  CONSTRAINT fk_attribution_reel FOREIGN KEY (reel_id) REFERENCES reels(reel_id) ON  
DELETE CASCADE,  
  CONSTRAINT fk_attribution_campaign FOREIGN KEY (campaign_id) REFERENCES  
campaigns(campaign_id) ON DELETE CASCADE,  
  CONSTRAINT fk_attribution_product FOREIGN KEY (product_id) REFERENCES  
products(product_id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.7 Alter order_items — add attributed_campaign_id

SQL

```
ALTER TABLE order_items ADD COLUMN attributed_campaign_id INT NULL AFTER merchant_id;  
ALTER TABLE order_items ADD KEY idx_order_items_attributed_campaign  
(attributed_campaign_id);  
ALTER TABLE order_items ADD CONSTRAINT fk_order_items_attributed_campaign FOREIGN KEY  
(attributed_campaign_id) REFERENCES campaigns(campaign_id) ON DELETE SET NULL;
```

13.1.8 Create settlement_ledger

SQL

```
CREATE TABLE IF NOT EXISTS settlement_ledger (
  id BIGINT NOT NULL AUTO_INCREMENT,
  order_item_id INT NOT NULL,
  campaign_id INT NOT NULL,
  creator_id INT NOT NULL,
  merchant_id INT NOT NULL,
  eligible_amount DECIMAL(12,2) NOT NULL,
  creator_commission_amount DECIMAL(12,2) NOT NULL,
  platform_fee_amount DECIMAL(12,2) NOT NULL,
  merchant_net_amount DECIMAL(12,2) NOT NULL,
  currency VARCHAR(3) NOT NULL DEFAULT 'INR',
  status VARCHAR(32) NOT NULL DEFAULT 'pending_payout',
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  paid_at DATETIME NULL,
  payout_id INT NULL,
  PRIMARY KEY (id),
  KEY idx_settlement_campaign (campaign_id),
  KEY idx_settlement_creator (creator_id),
  KEY idx_settlement_status (status),
  CONSTRAINT fk_settlement_order_item FOREIGN KEY (order_item_id) REFERENCES
order_items(order_item_id) ON DELETE CASCADE,
  CONSTRAINT fk_settlement_campaign FOREIGN KEY (campaign_id) REFERENCES
campaigns(campaign_id) ON DELETE CASCADE,
  CONSTRAINT fk_settlement_creator FOREIGN KEY (creator_id) REFERENCES
creator_profiles(id) ON DELETE CASCADE,
  CONSTRAINT fk_settlement_merchant FOREIGN KEY (merchant_id) REFERENCES
merchant_profiles(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.9 Create creator_payouts

SQL

```
CREATE TABLE IF NOT EXISTS creator_payouts (  
  id INT NOT NULL AUTO_INCREMENT,  
  creator_id INT NOT NULL,  
  amount DECIMAL(12,2) NOT NULL,  
  currency VARCHAR(3) NOT NULL DEFAULT 'INR',  
  status VARCHAR(32) NOT NULL DEFAULT 'pending',  
  period_start DATETIME NOT NULL,  
  period_end DATETIME NOT NULL,  
  payment_method VARCHAR(32) NULL,  
  reference_id VARCHAR(100) NULL,  
  paid_at DATETIME NULL,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  KEY idx_creator_payouts_creator (creator_id),  
  KEY idx_creator_payouts_status (status),  
  CONSTRAINT fk_creator_payouts_creator FOREIGN KEY (creator_id) REFERENCES  
creator_profiles(id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

13.1.10 Add payout_id to settlement_ledger (if not in 13.1.8)

SQL

```
ALTER TABLE settlement_ledger ADD CONSTRAINT fk_settlement_payout FOREIGN KEY  
(payout_id) REFERENCES creator_payouts(id) ON DELETE SET NULL;
```

13.1.11 campaign_code generationGenerate unique code in application: e.g. **CAMP-** + uppercase alphanumeric (e.g. 6 chars from timestamp + random), check uniqueness and retry on collision.

13.2 Campaign State Machine — Transition Table

Allowed transitions: (current_status × action × role) → new_status. Any other combination returns **409 Conflict** with code **INVALID_STATE**.

Current status	Action	Role	New status
draft	send	merchant	sent
draft	cancel	merchant	cancelled
sent	accept	creator	active
sent	reject	creator	rejected
sent	cancel	merchant	cancelled
active	cancel	merchant	cancelled
active	cancel	admin	cancelled
active	(creator submits reel)	creator	submitted *
submitted	approve_reel	merchant	live *
submitted	reject_reel	merchant	active

live	mark_complete	merchant / system	completed
—	expire (window_end < now)	system	completed

* Optional: keep campaign status **active** and only reel approval_status changes; or move campaign to **submitted** when first reel submitted and to **live** when first reel approved. Document choice in implementation.

13.3 RBAC Matrix (Role × Resource × Action)

Resource	Action	Creator	Merchant	Admin
own profile	create (onboarding)	✓	—	—
own profile	read, update	✓	—	✓ (view)
creator_categories	read, update (own)	✓	—	✓
creators (discovery)	list	—	✓	✓
campaigns	create (draft)	—	✓ (own)	—
campaigns	send, cancel, extend	—	✓ (own)	✓ (override)
campaigns	list	—	✓ (own)	✓ (all)
campaigns	get by id	—	✓ (own)	✓ (all)
campaigns (assigned to me)	list, get, accept, reject	✓	—	—
campaigns (assigned to me)	list active (for upload)	✓	—	—
reels (creator upload)	create (campaign-linked)	✓	—	—
reels (pending for my campaign)	list, approve, reject	—	✓ (own campaign)	✓
reels	read (feed, public)	✓	✓	✓
attribution_events	create (click)	—	—	—
attribution (click API)	call	user/guest	—	—
creator earnings	read own	✓	—	✓
merchant settlement	read (per campaign)	—	✓ (own)	✓ (all)
settlement ledger	read / run jobs	—	—	✓
creator payouts	read own	✓	—	✓
creator payouts	trigger run	—	—	✓

Resource-level rule: Creator can only access campaigns where `creator_id` = current user's `creator_profile.id`. Merchant can only access campaigns where `merchant_id` = current user's `merchant_profile.id`.

13.4 Configuration Specification

Key	Type	Default	Description	Used in
ATTRIBUTION_WINDOW_DAYS	int	7	Days within which last reel→product click attributes sale.	Attribution resolution
PLATFORM_FEE_PERCENT	decimal	5	AOIN fee % on attributed sales.	Settlement
ELIGIBILITY_RETURN_DAYS	int	14	Days after delivery after which item is eligible (align with order policy).	Eligibility job
CAMPAIGN_CODE_PREFIX	string	CAMP-	Prefix for <code>campaign_code</code> .	Campaign create
CAMPAIGN_CODE_LENGTH	int	6	Alphanumeric length after prefix.	Campaign create
REEL_VIDEO_MAX_SIZE_MB	int	100	Max upload size in MB.	Reel upload
REEL_VIDEO_ALLOWED_EXTENSIONS	list	mp4, webm, mov	Allowed video extensions.	Reel upload
CREATOR_MODULE_ENABLED	bool	true	Feature flag for creator module.	All creator/merchant creator APIs
PAYOUT_CURRENCY	string	INR	Default currency for ledger and payouts.	Settlement, creator_payouts

13.5 Error Codes Reference

Code	HTTP	Description	When
VALIDATION_ERROR	400	Request body/query invalid.	Missing/invalid fields, min 5 categories, etc.
INVALID_STATE	400	State transition not allowed.	e.g. Send from non-draft, Accept from non-sent.
UNAUTHORIZED	401	Missing or invalid token.	No JWT or expired.
FORBIDDEN	403	Valid token but role/resource not allowed.	Merchant calling creator API, wrong campaign owner.
NOT_FOUND	404		

		Resource not found or not accessible.	Campaign/reel/profile not found or not owned.
CONFLICT	409	Conflict with current state (e.g. already exists).	Onboarding already done, duplicate accept.
PAYLOAD_TOO_LARGE	413	Request entity too large.	Reel video exceeds max size.
INTERNAL_ERROR	500	Unexpected server error.	Log and return generic message.

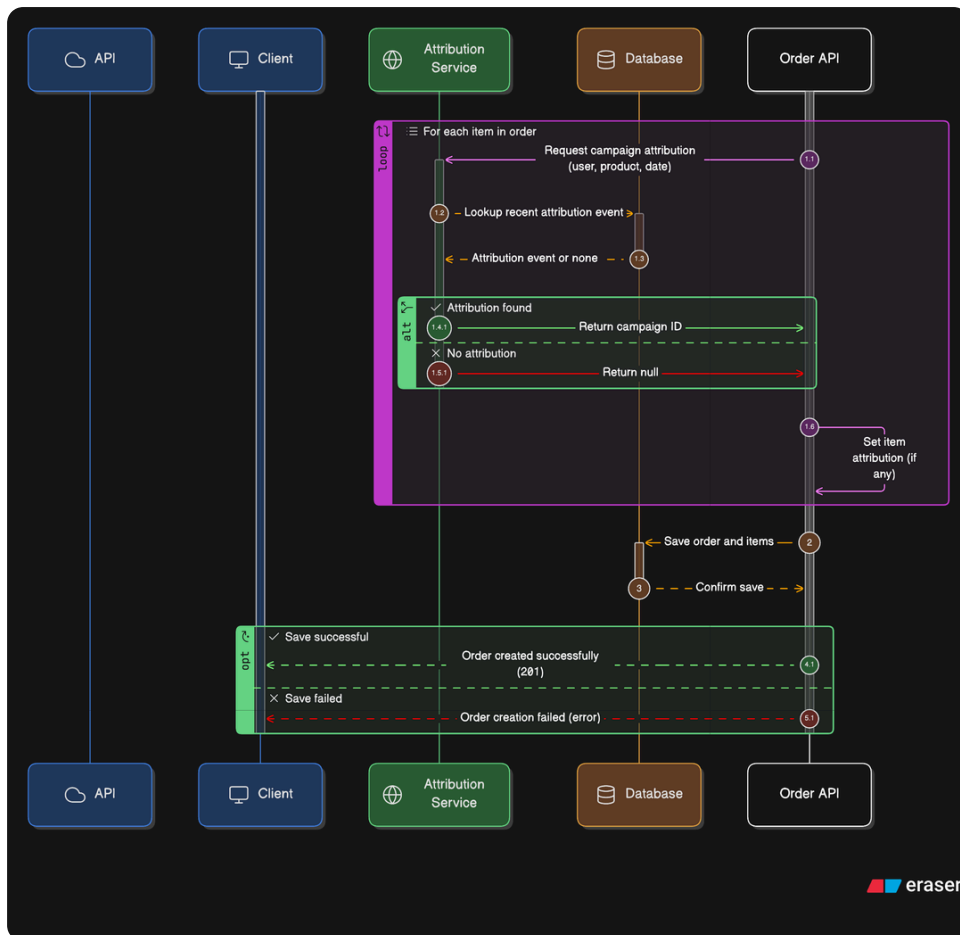
All error responses: { "error": "<message>", "code": "<CODE>", "details": { ... } }.

13.6 Sequence Diagrams (Key Flows)

13.6.1 Attribution resolution (synchronous at order placement)

Y Mermaid Diagram

```
sequenceDiagram
    participant Client
    participant OrderAPI
    participant AttributionSvc
    participant DB
    Client->>OrderAPI: POST /orders (create order)
    OrderAPI->>OrderAPI: Create order + order_items (product_id, quantity, ...)
    loop For each order_item
        OrderAPI->>AttributionSvc: resolve_last_touch(user_id, product_id, order_date)
        AttributionSvc->>DB: SELECT from attribution_events WHERE user_id, product_id, clicked_at >= order_date - X days ORDER BY clicked_at DESC LIMIT 1
        DB-->>AttributionSvc: event (campaign_id) or null
        AttributionSvc-->>OrderAPI: campaign_id or null
        OrderAPI->>OrderAPI: Set order_item.attributed_campaign_id
    end
    OrderAPI->>DB: Commit order + items
    OrderAPI-->>Client: 201 Order created
```



13.6.2 Eligibility job (async)

Mermaid Diagram

sequenceDiagram

participant Cron

participant EligibilityJob

participant DB

Cron->>EligibilityJob: Run (e.g. daily)

EligibilityJob->>DB: Find order_items with attributed_campaign_id, order delivered, delivery_date + ELIGIBILITY_RETURN_DAYS < now, not yet in settlement_ledger

DB-->>EligibilityJob: List of order_item_idVs

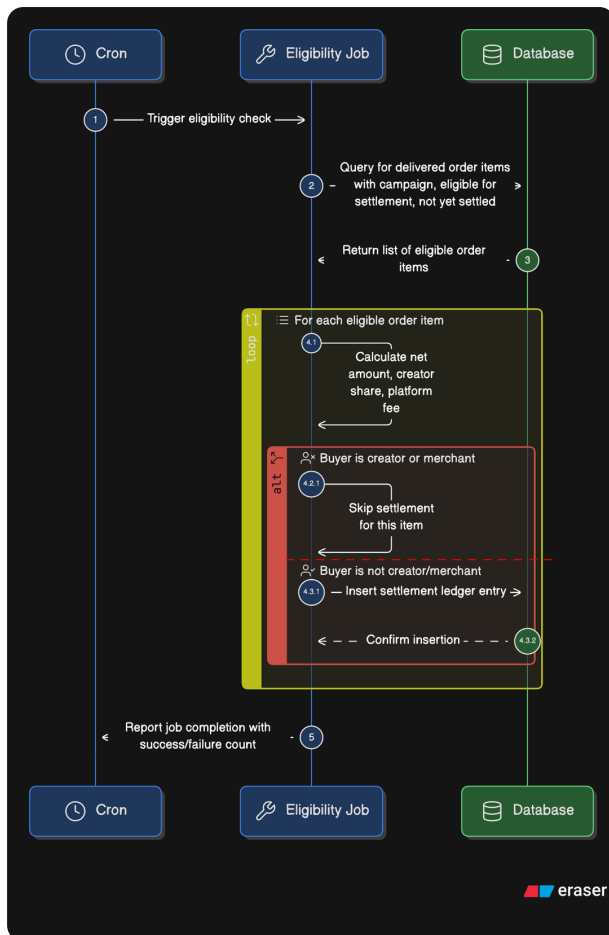
loop For each item

EligibilityJob->>EligibilityJob: Compute net, creator %, platform fee;
skip if buyer = creator/merchant

EligibilityJob->>DB: INSERT settlement_ledger

end

EligibilityJob-->>Cron: Done (log success/failure count)



13.6.3 Payout run (async)

Mermaid Diagram

sequenceDiagram

participant Admin

participant PayoutJob

participant DB

participant Bank

Admin->>PayoutJob: Trigger (e.g. POST /api/admin/settlement/run-payout)

PayoutJob->>DB: SELECT creator_id, SUM(creator_commission_amount) FROM settlement_ledger WHERE status = 'pending_payout' GROUP BY creator_id

DB-->>PayoutJob: Per-creator totals

loop For each creator

PayoutJob->>DB: INSERT creator_payouts (pending)

PayoutJob->>Bank: Initiate transfer (or export CSV)

alt Success

PayoutJob->>DB: UPDATE settlement_ledger SET status='paid', paid_at, payout_id

PayoutJob->>DB: UPDATE creator_payouts SET status='completed', paid_at, reference_id

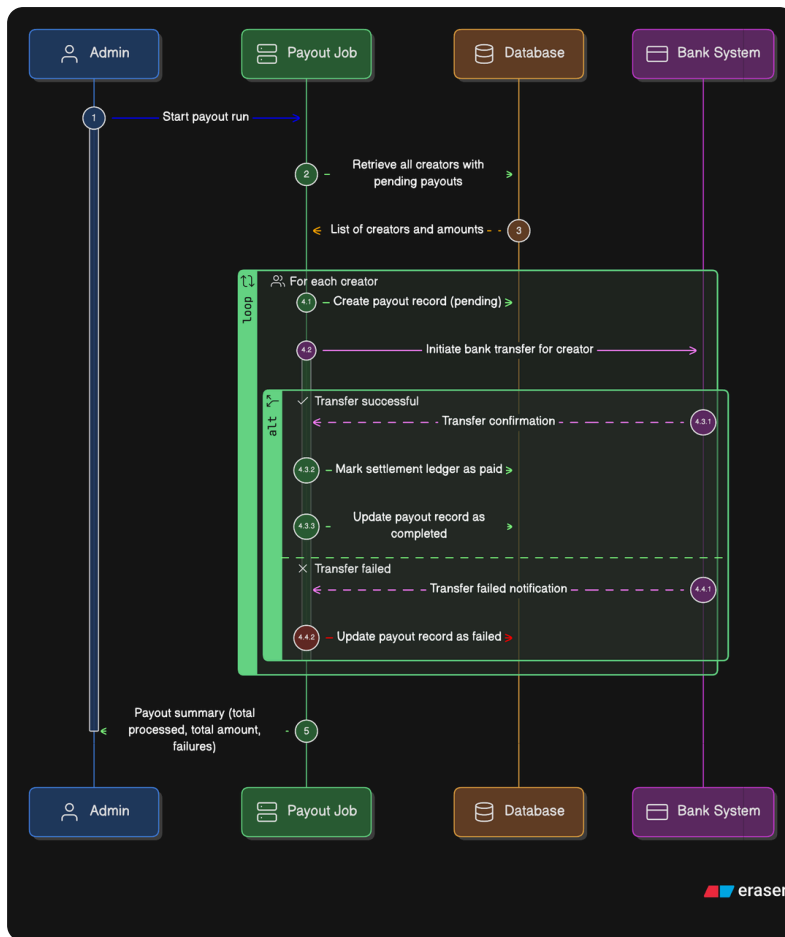
else Failure

PayoutJob->>DB: UPDATE creator_payouts SET status='failed'

end

end

PayoutJob-->>Admin: Summary (count, total amount, failures)



13.7 Test Scenarios (Key Flows)

#	Scenario	Steps	Expected
T1	Creator signup and onboarding	1) POST signup role=creator 2) POST onboarding (5 categories, availability)	201 both; profile and categories created.
T2	Merchant creates and sends campaign	1) GET creators (filter category) 2) POST campaigns (draft) 3) POST campaigns/:id/send	200 list; 201 campaign; 200 sent; status=sent.
T3	Creator accepts campaign	1) GET creator/campaigns (status=sent) 2) POST creator/campaigns/:id/accept	200 list; 200 accept; status=active.
T4	Creator uploads reel	1) GET creator/campaigns/active 2) POST creator/reels (campaign_id, video, description)	200 list; 201 reel; approval_status=pending.
T5			

	Merchant approves reel	1) GET campaigns/:id/reels/pending 2) POST reels/:id/approve	200 list; 200 approve; reel visible in feed.
T6	Attribution and order	1) POST attribution/click (reel_id, product_id) as user 2) POST orders (cart with that product)	201 click; 201 order; order_item.attributed_campaign_id set.
T7	Eligibility and ledger	1) Set order to delivered; advance delivery date + 14 days 2) Run eligibility job	settlement_ledger rows created for attributed items; status=pending_payout.
T8	Self-order no attribution	1) Creator clicks product from own reel 2) Creator places order	attribution event stored; at order placement attributed_campaign_id NOT set (buyer=creator).
T9	Invalid state rejected	POST campaigns/:id/send twice; POST creator/campaigns/:id/accept for already accepted campaign	400/409 second call; status unchanged.
T10	RBAC: creator cannot see other creator campaigns	GET creator/campaigns with campaign for another creator	404 or empty for that id.

13.8 Rollback & Failure Handling

Situation	Action	Rollback / Mitigation
Migration fails mid-way	Stop; fix script or data; re-run from last successful step.	Do not run application with partial schema; add FKs only after referenced tables exist.
Deploy new code without migration	New code expects new columns/tables.	Deploy migrations first, then app; or feature-flag new paths until migration run.
Eligibility job fails	Log error; alert; retry with backoff.	Job idempotent (skip already-settled items); safe to re-run.
Payout run fails for one creator	Mark that payout as failed; continue others; alert.	Retry single creator later; do not double-pay (payout record created with status=failed).
Attribution write fails	Return 500 to client; client may retry.	Optionally accept idempotent by (user, reel, product, time bucket) to dedupe.
	Orphan object in S3.	

Reel upload fails after S3 write		Periodic cleanup job for unreferenced keys; or rollback S3 delete on DB failure.
Campaign state inconsistent	Admin override via PATCH admin/campaigns/:id.	Document allowed overrides; audit log.
Data fix required	Prefer script with transaction; backup first.	Restore from backup if script wrong; run fix in transaction.

13.9 API Changelog (Placeholder)

Version	Date	Change
v1	2026-02-24	Initial Creator module APIs per CREATOR_MODULE_API_SPEC.md (./CREATOR_MODULE_API_SPEC.md).

When adding or changing APIs, document here: new endpoints, new request/response fields, breaking changes, deprecations.

Document History & References

Version	Date	Change
1.0	2026-02-24	Initial PRD: current state analysis, target state, gaps, flows, data models, APIs, phases.
1.1	2026-02-24	Added Section 6: UI/UX Sections & Screen Maps (diagrams for Creator, Merchant, Customer, Admin). Renumbered Sections 7–11.
1.2	2026-02-24	Added Section 7.3–7.5: Entity-Relationship diagram, Table Connectivity Matrix, and DFDs (Level 0 context, Level 1 processes, Level 2 attribution & settlement).
1.3	2026-02-24	Expanded Section 8: detailed API spec with expected request/response for 8.1 (signup, login); added CREATOR_MODULE_API_SPEC.md with every Creator module API and full req/res; added API Quick Reference Table.
1.4	2026-02-24	Added Section 12: Architecture Gaps & Missing Specifications — Software Architect review: NFRs, security/RBAC, configuration, integration, data/compliance, concurrency, attribution details, settlement/payout, data model gaps, observability, deployment, testing, role/identity, lifecycle edge cases, documentation.
1.5	2026-02-24	

	Added Section 13: Implementation Reference — DDL/migration snippets (MySQL) for all new/changed tables; Campaign state machine transition table; RBAC matrix (Role × Resource × Action); Configuration specification (keys, type, default); Error codes reference; Sequence diagrams (attribution resolution, eligibility job, payout run); Test scenarios (T1–T10); Rollback & failure handling.
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References:

- CREATOR_INFLUENCER_MODULE_HINGLISH_OVERVIEW.md
(./CREATOR_INFLUENCER_MODULE_HINGLISH_OVERVIEW.md) — Source requirements (Hinglish).
- Codebase: `auth/models/models.py` (User, UserRole, MerchantProfile), `models/reel.py`, `controllers/reels_controller.py`, `models/order.py`, `models/shipment.py`, `models/enums.py`, `models/visit_tracking.py`.

End of PRD.